

Harmonizing Progress and Privacy: A Framework for an Indian AI Act in Healthcare

Chapter 1: The Twin Imperatives of the Digital Age: Data Protection and AI-Driven Efficiency

1.1 The Sanctity of Personal Data: Foundational Principles and the Need for Protection

In an era defined by the pervasive collection, analysis, and monetization of information, the protection of personal data has emerged as one of the most pressing legal and ethical issues. Every digital interaction, from accessing healthcare portals to using social media, generates a data footprint that can be leveraged by corporations, governments, and malicious actors [[Aidun, 2025, *Data Privacy in the Digital Age: A Comparative Analysis of U.S. and EU Regulations*, 2](#)]. This reality has given rise to a global consensus on the need for robust legal frameworks to govern the processing of personal information, recognizing data privacy not as a luxury but as a fundamental human right essential for individual autonomy, dignity, and trust in the digital ecosystem. [[IEEE Publication, 2023, *What Is Digital Privacy and Its Importance?* 77](#)]

The digital transformation of healthcare has created vast pools of sensitive personal data, heightening privacy concerns. Data protection laws seek to give individuals control over their information. For example, India's highest court recognized privacy as a fundamental right, prompting comprehensive data rules. [[DLA Piper, 2024, *Data protection laws in India* 44](#)] Globally, regulations like the EU's GDPR codify principles of consent, transparency and security. Such laws rest on fundamental rights: as one commentary notes, protecting personal data is part of the right to a private life. [[Boodhonee-Aikat and Virahsawmy-Naik, 2018 *Mauritius Updates Its Data Protection Legislation To Be In Line With GDPR*, 19](#)] These protections guard against harms like identity theft or discrimination that can arise from health data misuse.

Robust privacy frameworks also help build trust, which is vital in healthcare. Patients are more likely to share data for research or AI-driven care if they trust their data are safe. In the absence of regulation, data breaches are common – for example, in 2023 the US healthcare sector saw hundreds of reported breaches exposing millions of patient records. [[HIPAA Journal](#) – Steve Alder, 2025 *Healthcare Data Breach Statistics*, 5] Data protection laws aim to prevent such outcomes by requiring encryption, breach notification, and limits on data collection and use. By setting a legal floor for privacy, they ensure that innovation in healthcare AI does not override individuals' rights.

Modern data protection laws are built upon a set of core principles that have been harmonized across international standards, most notably articulated in the European Union's General Data Protection Regulation (GDPR) and echoed in frameworks by the United Nations (UN) and the World Bank. These foundational principles serve as the bedrock for responsible data handling. First, the principle of **Lawfulness, Fairness, and Transparency** [[UN System Chief Executives Board for Coordination, 2024, 154](#)] mandates that data processing must have a valid legal basis, must not be unduly detrimental or misleading to individuals, and must be communicated clearly and honestly. This ensures that individuals are aware of how their data is being used and for what reasons.

Second, **Purpose Limitation** dictates that personal data shall be collected for specified, explicit, and legitimate purposes and not be further processed in a manner incompatible with those initial purposes. This prevents "function creep," where data collected for one reason is later repurposed for another without the individual's knowledge or consent. Closely related is the principle of [[The World Bank Practitioner's Guide on Data protection and privacy laws](#), Section III, 2021, 161] **Data Minimization**, which requires that data collection be limited to what is adequate, relevant, and necessary for the stated purpose. Organizations are prohibited from gathering "extra" data in the hope that it might become useful later. [[IT Governance Report 82](#)]

Furthermore, the principles of **Accuracy** and **Storage Limitation** obligate organizations to ensure the data they hold is correct and up-to-date, and that it is not kept for longer than necessary to fulfill its purpose [[The World Bank Practitioner's Guide on Data protection and privacy laws](#), Section III, 2021, 161]. The principle of **Integrity and Confidentiality** requires the implementation of appropriate technical and organizational security measures, such as encryption and access controls, to protect data from unauthorized access, corruption, or loss. Finally, the principle of **Accountability** holds organizations responsible for demonstrating compliance with all of these principles, often through measures like maintaining records of processing activities, appointing a Data Protection Officer (DPO), and conducting Data Protection Impact Assessments (DPIAs). The imperative for these legal safeguards is magnified when considering sensitive personal data, particularly health information. The unauthorized exposure or misuse of health data can lead to severe consequences, including discrimination, financial loss, and significant personal harm. Therefore, the application of these principles in the healthcare context is not merely a matter of legal compliance but a prerequisite for maintaining patient trust and ensuring the ethical delivery of care in an increasingly digital world. [[IEEE Publication, 2023, What Is Digital Privacy and Its Importance?](#) 77]

1.2 The Promise of Artificial Intelligence in Healthcare: A Paradigm Shift in Efficiency and Efficacy

Concurrent with the rise of data privacy as a societal imperative, artificial intelligence (AI) has emerged as a transformative force with the potential to revolutionize nearly every facet of healthcare. By harnessing vast amounts of data, advanced algorithms, and machine learning capabilities, AI promises to address some of the most persistent challenges in the sector: rising costs, operational inefficiencies, and the ever-increasing demand for higher-quality, personalized care. Healthcare is increasingly data-driven, making privacy protection crucial. [[Boodhonee-Aikat and Virahsawmy-Naik, 2018](#) *Mauritius Updates Its Data Protection Legislation To Be In Line With GDPR*, 19] notes “the need for data protection law is derived from the principle that everyone has the right to the protection of their private life, of which personal data forms an integral part”. At the same time, artificial intelligence (AI) promises to revolutionize healthcare by improving diagnostics, resource allocation, and patient care. The WHO emphasizes that “AI is already playing a role in diagnosis and clinical care, drug development, disease surveillance...The future of healthcare is digital, and we must do what we can to promote universal access to these innovations”. [1] AI can “improve access to services, address workforce shortages, and reduce health system costs”. [[Seventy-seventh World Health Assembly Strategic Roundtable](#), 2024, 147] These dual imperatives protecting individual privacy and fostering AI innovation – must be balanced in health policy.

One of the most immediate and impactful applications of AI is in enhancing **operational and administrative efficiency**. Healthcare systems are notoriously burdened by administrative tasks, with physicians often spending a significant portion of their time on electronic health records (EHRs) and desk work rather than direct patient care. AI can automate many of these routine, repetitive processes, such as patient appointment scheduling, billing, insurance claims processing, and data entry into EHRs. By deploying AI for these functions, healthcare organizations can significantly reduce administrative costs, which account for a substantial portion of healthcare spending, minimize human error, and free up clinicians to focus on more complex, patient-facing responsibilities. This shift not only improves productivity but also helps mitigate the growing problem of physician burnout.

Beyond administration, AI is making profound contributions to **clinical and diagnostic enhancement**. AI algorithms, particularly those based on deep learning, have demonstrated a remarkable ability to analyze complex medical data, often with a level of accuracy and speed that matches or exceeds human experts [[Davenport and Kalakota, 2019](#), *The potential for artificial intelligence in healthcare*, 38]. In medical imaging, AI tools are being used to analyze X-rays, CT scans, and MRIs to detect early signs of diseases like cancer, fractures, or diabetic retinopathy, identifying subtle abnormalities that might be missed by the human eye. [[Park University, 2024](#), *AI in Healthcare: Enhancing Patient Care and Diagnosis*, 122] This leads to earlier and more accurate diagnoses, which is critical for improving patient outcomes. [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52] Furthermore, AI-powered predictive analytics can continuously monitor patient data from various sources to forecast the onset of critical conditions like sepsis or cardiac arrest hours

before clinical symptoms appear, enabling timely and life-saving interventions. [[Artificial intelligence in healthcare - European Parliament](#), 2022, 53]

Finally, AI is a key enabler of **personalization and accelerated drug discovery**. By analyzing a patient's unique genetic makeup, lifestyle data, and medical history, AI can help develop tailored treatment plans, moving healthcare away from a "one-size-fits-all" model toward precision medicine [[Park University, 2024](#), *AI in Healthcare: Enhancing Patient Care and Diagnosis*, 122] This personalized approach can improve the effectiveness of therapies while reducing adverse side effects. In the pharmaceutical realm, AI is drastically accelerating the drug discovery and development pipeline. It can analyze vast datasets of chemical compounds to predict which are most likely to be effective against a particular disease, simulate clinical trials, and identify suitable patient candidates, significantly reducing the time and billions of dollars traditionally required to bring a new drug to market.

In healthcare, AI is applied to diagnostics, monitoring, and even drug development. European Commission reports illustrate AI applications like early sepsis detection (predicting life-threatening conditions in intensive care) and breast cancer screening (AI systems surpassing human radiologists in some cases) [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52]. A concrete example is a homecare optical coherence tomography (OCT) device for eye care: elderly patients take daily eye scans, which are sent to a cloud AI for analysis, alerting their ophthalmologist to intervene if needed [[Bende et. al 2022](#), *A Case Study on Data Protection for a Cloud- and AI-based Homecare Medical Device*, 17]. Such systems require collecting sensitive health data (imaging and health records) and rely on cloud connectivity.

Health-related data are typically gathered from electronic health records (EHR), imaging devices (MRI, Xray), wearable sensors (heart rate, activity trackers), genomics, and patient-generated data. Large labeled datasets (e.g. NIH X-ray dataset, private clinical archives) are used to train models, often in the cloud or on shared research databases. As Chand and others note, AI in medicine demands “much larger volumes of data” than traditional telemedicine, often stored in multi-institutional repositories [[Yadav et al 2023](#), *Data Privacy in Healthcare: In the Era of Artificial Intelligence*, 165]. For instance, the Cancer Imaging Archive (TCIA) and mammography datasets like DDSM are publicly used to develop diagnostic algorithms. In practice, data can come from hospital systems, networked devices, or even consumer apps syncing with clinicians.

These data collection practices raise privacy challenges. Medical data are treated as highly sensitive: GDPR classifies health as a “special category” requiring extra safeguards (explicit consent, or processing only for health care/legal reasons). The DPDP Act, by contrast, does not segregate sensitive categories – all personal data (only if digital) are generally covered, but with no special health carve-outs beyond the normal consent regime [[DPDP Act 2023, Section 2\(t\) & \(n\), Section 3, Meity](#), 45]. In Europe, processing of health data for healthcare purposes is allowed under GDPR Art. 9 (b)-(c) if done by professionals, but commercial AI projects often partner private firms with clinics, complicating legal bases. This was starkly illustrated by the 2016 DeepMind–NHS partnership: Google’s AI team accessed over a million NHS patient records without proper patient consent, and regulators later found “inappropriate legal basis” for this data sharing [[Murdoch 2021](#), *Privacy and artificial intelligence: challenges for protecting health information in a new era*, 112]. The transfer of those records out of the UK to the USA also exposed cross-jurisdictional loopholes.

Human–AI confusion is evident in clinics too. AI models can be opaque and make errors that human doctors might not anticipate. [[Goddard et al. 2011](#), *Automation bias: a systematic review of frequency, effect mediators, and mitigators*, 62] reviewed “automation bias” in clinical decision support: clinicians using an AI system tended to over-rely on its output, sometimes accepting incorrect AI advice and thus introducing new errors. For example, if a CDSS suggests a diagnosis, a physician under pressure may follow it even when skeptical, committing an error of commission. Conversely, an AI system might fail to flag a critical detail, leading to omission errors. This shows how an AI’s “decision” can conflict with a human doctor’s reasoning, causing confusion in clinical judgment. The “black box” nature of many models compounds this: because AI “reasoning” is opaque, even the data used can be unclear. [[Murdoch 2021](#), *Privacy and artificial intelligence: challenges for protecting health information in a new era*, 112] warns that novel algorithms now can re-identify supposedly anonymized medical

data, and that health AI often cannot be fully supervised by humans due to this opacity. Thus, both clinicians and patients may struggle to interpret AI outputs without clear explanations.

The effective deployment of AI in healthcare, however, presents a fundamental paradox when viewed through the lens of data protection. The very principles designed to protect individuals, such as data minimization and purpose limitation, stand in direct tension with the operational requirements of AI systems, which thrive on vast and diverse datasets to achieve accuracy and fairness [[IEEE Publication, 2023, What Is Digital Privacy and Its Importance?](#) 77] A regulatory framework that rigidly enforces data minimization could inadvertently cripple the development of robust AI, while unfettered data collection would violate fundamental privacy rights. This inherent conflict necessitates a new, more nuanced approach to governance that can balance these competing imperatives.

Furthermore, the most ethically sound and practically effective applications of AI in medicine position it as a tool to *augment* human intelligence, not replace it. [[Hitachi Digital Services, 2024, AI-Fueled Productivity: AI Opens New Era of Efficiency in Healthcare](#), 71] AI excels at data-intensive analysis and automation, but it lacks the empathy, critical judgment, and holistic understanding of a human clinician. [[TOI Report, 2025, You and AI: Is Dr AI good for your health?](#), 151] The optimal model is one of "augmented intelligence," where AI provides powerful insights and efficiencies, but the final clinical decision-making and patient relationship remain firmly in human hands. This understanding implies that any regulatory framework must mandate "human-in-the-loop" oversight for high-risk applications, ensuring AI serves as a trusted co-pilot, not an autonomous pilot, in the journey of patient care.

Chapter 2: The Human Prerequisite for Advanced AI: A Case for Investing in Human Capital

2.1 Education as the Bedrock of Ethical Innovation and Technological Adoption

The advancement of sophisticated technologies like artificial intelligence is inextricably linked to the quality of a nation's human capital. A well-educated populace is not merely a passive consumer of technology but the fundamental prerequisite for its development, adoption, and responsible integration into society. Academic research consistently demonstrates that human capital, particularly when enhanced through education, is a primary driver of technological diffusion and economic growth [[Cosar 2011, Human Capital, Technology Adoption and Development](#), 36]. Education improves the quality of the workforce, which in turn boosts a nation's innovative capacity. [[Bawono 2021, Human Capital, Technology, and Economic Growth: A Case Study of Indonesia](#), 16] In the context of AI, this relationship is twofold. First, a skilled workforce, especially in STEM fields, is necessary to build, deploy, and maintain complex AI systems. [[Reenen 2021, INNOVATION AND HUMAN CAPITAL POLICY](#), 138] Second, a broadly educated society is better equipped to understand the implications of AI, engage in informed public discourse, and adapt to the economic transformations it brings.

However, the role of education extends beyond technical proficiency into the critical domain of ethics. The development of trustworthy AI systems that are fair, transparent, accountable, and safe actually depends on creators who possess not only technical skills but also a strong ethical grounding [[Center for Teaching Innovation - Cornell University, Ethical AI for Teaching and Learning](#), 23]. Educational institutions have a pivotal role in shaping this next generation of AI professionals by integrating ethics directly into their curricula. Courses that encourage critical thinking about bias mitigation, data privacy, accountability, and the societal impact of AI are essential. By fostering interdisciplinary collaboration between computer science, law, philosophy, and social sciences, universities can equip students to address the multifaceted ethical challenges posed by AI. In this sense, education acts as a societal safeguard, cultivating a generation of innovators who understand that the central question is not just *if* we can build a technology, but *if* and *how* we should. This ethical awareness is fundamental to preventing the misuse of AI and ensuring that its development aligns with human values and fundamental rights.

2.2 The Symbiotic Relationship Between Healthcare, Human Capital, and AI Quality

The argument for investing in human capital can be extended into a more nuanced and powerful thesis: strategic investment in public health is a direct and indispensable investment in a nation's technological capacity. A healthy population is the foundation of productive and innovative human capital. [[Developing Human Capital to Support U.S. Innovation Capacity – A Workshop, 2023, 116](#)] This connection is particularly profound in the context of healthcare AI, where the quality of the system is a direct reflection of the quality of the human ecosystem that creates and trains it.

This symbiotic relationship manifests throughout the AI lifecycle. First, healthy and well-educated individuals are the innovators who conceive of and build advanced AI systems. A nation cannot lead in a knowledge-intensive field like AI without a robust talent pipeline, which is itself dependent on the health and well-being of its citizens. More critically, however, is the role of the healthcare system in generating the data that serves as the lifeblood of AI models. AI algorithms learn from the data they are fed, and if that data is flawed, the AI's outputs will be flawed.

The problem of AI bias is not, at its core, a technological failure; it is a mathematical reflection of pre-existing human and systemic biases present in our society and its institutions. [Tarasiuk, 2025, [Forbes Human Resources Council](#), *AI Bias In Hiring Is An HR Problem, Not A Tech Issue*, 149] When a healthcare system is plagued by inequalities, overworked staff, and poor data collection practices, the data it produces will be inherently biased. For example, if certain demographic groups historically receive less medical attention or are underrepresented in clinical trials, an AI trained on this data will learn these disparities as normal patterns and perpetuate, or even amplify, them in its recommendations. [[Chapman University AI Hub](#), 26] A prominent case study is the commercial algorithm used in U.S. hospitals that systematically underestimated the health needs of Black patients because it used healthcare cost as a proxy for illness, failing to account for the fact that less money is historically spent on Black patients with the same conditions. [Weber, 2023, [Hopkins Bloomberg Public Health Magazine](#), *Rooting Out AI's Biases*, 159] The algorithm, in effect, learned and automated a societal bias, reducing the care Black patients received by over 50%.

As a result, healthcare investment is the foundation for AI. A recent global health analysis warns that “AI cannot be deployed effectively or ethically in contexts lacking sufficient... infrastructure and resources” [[McCoy 2025](#), *Building health systems capable of leveraging AI: applying Paul Farmer's 5S framework for equitable global health*, 103]. In under-resourced health systems, pouring funds into AI without basic services risks waste and harm. Thus, strengthening primary care, training, and equipment is a prerequisite. When hospitals have reliable data, connectivity, and trained staff, then AI tools (e.g. remote diagnostics) can truly improve outcomes. In short, human capital i.e. educated citizens and a robust health system is needed to reap AI's promised gains without compromising quality or equity.

This reveals a critical insight: to create the high-quality, representative, and unbiased datasets required for the next generation of trustworthy healthcare AI, a society must first invest in the human systems that generate that data. Improving the quality of and access to healthcare, training medical professionals to recognize and combat their own implicit biases, and ensuring equitable resource distribution are not just social goals, they are essential data cleansing and quality control measures for the AI era. Therefore, public health expenditure should be reframed as a long-term strategic investment that cultivates a virtuous cycle. Better healthcare creates healthier, more capable human capital. This improved human capital, in turn, generates higher quality data and produces more skilled developers, leading to the creation of more accurate and fair AI tools. These superior AI tools can then be deployed back into the healthcare system to further enhance efficiency and patient outcomes, completing the cycle and driving both societal well-being and technological leadership. The presence of bias in an AI algorithm should thus be treated as a powerful diagnostic tool, providing a quantitative signal of underlying systemic failures in the real-world healthcare system that require both technical and policy-level interventions.

Chapter 3: Global Paradigms in AI Governance: A Comparative Legal Analysis

3.1 The European Union's Rights-Centric Model: GDPR and the AI Act

The European Union has established itself as a global leader in digital regulation through a comprehensive, rights-centric approach. Its governance model for artificial intelligence is built upon two foundational pillars: the General Data Protection Regulation (GDPR) and the EU AI Act. The GDPR serves as the bedrock, providing a robust, horizontal framework for the protection of personal data [[IAPP Report](#), *Top 10 operational impacts of the EU AI Act*, 76]. Enacted in 2018, it applies whenever the personal data of individuals in the EU is processed, regardless of whether AI is involved. Its principles of lawfulness, fairness, transparency, data minimization, and accountability, along with strong data subject rights, create a high standard of privacy protection that any AI system processing personal data must adhere to.

Modern data laws impose various requirements on AI in these sectors. Under the EU GDPR, data controllers in education or healthcare must ensure lawfulness, fairness, and transparency. Personal data (student or patient) can only be processed for specified purposes. Educational use often relies on consent (or legitimate interest), but for minors GDPR mandates parental consent and prohibits “profiling” without strong justification. In healthcare, GDPR’s Article 9 prohibits processing health data except under 10 strict conditions (e.g. explicit consent, medical necessity, public health) as outlined by the ICO [[Article 9, GDPR](#), 12]. Both domains must honor data subject rights: access, rectification, erasure, portability, and notably the right not to be subject to solely automated decisions. This “automated decision” right (Art. 22 GDPR) means a school or hospital cannot base important decisions only on an AI’s output (without human intervention) unless special conditions are met. In education, automated scoring of exams or admissions by AI would trigger this protection. In healthcare, patients could arguably invoke Art. 22 to refuse a fully automated diagnosis, though the interplay with medical device laws is debated [[Murdoch 2021](#), *Privacy and artificial intelligence: challenges for protecting health information in a new era*, 112]. GDPR also enforces data minimization and retention limits, which can clash with AI’s desire for rich datasets.

Building upon this foundation, the EU AI Act introduces a specific, cross-sectoral regulatory layer for AI systems themselves. [[EU AI Act – European Parliament](#), 2023, 54] Rather than a one-size-fits-all approach, the Act employs a risk-based classification system, categorizing AI applications into four distinct tiers: [[Deloitte Report](#), *Unpacking the EU AI Act: The future of AI governance*, 39]

Unacceptable Risk: Systems that pose a clear threat to the safety, livelihoods, and rights of people are banned outright. This includes AI used for social scoring by governments or manipulative subliminal techniques.

High Risk: This category includes AI systems used in critical sectors where they can have a significant impact on individuals' fundamental rights and safety. In practice, EU countries are already integrating AI within their healthcare frameworks under the new laws. For example, the EU’s Artificial Intelligence Act explicitly classifies medical AI software as “high risk,” triggering stringent requirements (risk mitigation, human supervision, quality controls) [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52]. Most healthcare AI applications, such as those used for medical diagnosis, treatment planning, or as safety components of medical devices, fall into this category. These systems are subject to stringent obligations, including rigorous conformity assessments before market entry, high-quality data governance, technical documentation, transparency, human oversight, and robust post-market monitoring. [[Lovells News Article](#), 2025, *AI Health Law & Policy: AI Health Law & Policy: Comparing regulatory landscapes for AI in medical devices in the EU and U.S.*, 73] Certain AI uses deemed unacceptable (e.g. manipulative social scoring, indiscriminate surveillance) are outright banned. [[Investopedia Article](#), Fallon-Peek, 2025, *What’s Inside the EU AI Act—and What It Means for Your Privacy*, 55] The goal is “trustworthy AI”: the Act aims to foster innovation while safeguarding safety and fundamental rights [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52].

Limited Risk: AI systems like chatbots are subject to specific transparency obligations, ensuring that users are aware they are interacting with a machine.

Minimal Risk: The vast majority of AI applications, such as AI-enabled video games or spam filters, fall into this category and are largely free from additional regulation. [[Deloitte Report](#), *Unpacking the EU AI Act: The future of AI governance*, 39]

The EU also launched initiatives like the European Health Data Space to share health data securely for AI research, combining innovation with privacy protections. Some EU data regulators have begun scrutinizing AI: e.g. Italy's privacy authority temporarily banned ChatGPT over GDPR concerns, showing enforcement of privacy rules on AI. These cases illustrate the EU system at work: AI innovation is encouraged, but within a framework designed to enforce GDPR rights and AI safety together.

Crucially, the EU's approach ensures the AI Act complements existing data protection rules. The Act explicitly states it is "without prejudice to" the GDPR [[Clark et. al. 2024](#), *Europe: The EU AI Act's relationship with data protection law: key takeaways*, 30]. As one analysis explains, the AI Act is a product safety law (governing system design) and does not replace GDPR's individual privacy rights. In practice, this means any AI processing personal health data must satisfy both frameworks: developers follow the EU Act's risk rules and GDPR's requirements on consent, minimization, and data subject rights. By design, the EU regime "works hand-in-glove", using AI regulation for systemic safeguards while relying on GDPR for personal privacy [[Clark et. al. 2024](#), *Europe: The EU AI Act's relationship with data protection law: key takeaways*, 30].

The interplay between the GDPR and the AI Act creates a dual-layered compliance environment for healthcare AI in the EU. An AI diagnostic tool, for example, must comply with the GDPR regarding the processing of patient health data and simultaneously meet the high-risk requirements of the AI Act concerning its safety, accuracy, and reliability. This comprehensive model prioritizes the protection of fundamental rights and aims to build public trust in AI [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52]. However, this approach has faced criticism from some industry stakeholders who argue that the heavy compliance burden, particularly for high-risk systems, could be "toxic" for innovation, slow down development, and place European companies at a competitive disadvantage compared to their counterparts in less regulated jurisdictions.

3.2 The United States' Market-Driven, Sector-Specific Approach

In contrast to the EU's centralized, horizontal framework, the United States has adopted a decentralized, market-driven, and sector-specific approach to AI governance. There is no single, comprehensive federal law analogous to the GDPR or the EU AI Act. Instead, AI regulation in the U.S. is a complex patchwork of existing laws, actions by federal agencies, and influential but voluntary frameworks.

In the U.S., implementation is more decentralized. Several states have passed or proposed laws affecting healthcare AI. For instance, Virginia's upcoming law requires healthcare entities to analyze algorithmic bias and allow clinicians to override AI recommendations (upon notifying patients) [[IAPP Article](#), Cole and Rolecki 2024, *Welcome to the 'Mild West': Balancing AI innovation, privacy principles*, 34]. In practice, major health systems have also self-regulated: some hospitals use AI "watch dogs" to audit AI tools before adoption. Yet without a unified mandate, these measures vary by institution. The U.S. experience shows that when data protection relies on sectoral laws (HIPAA, state privacy acts) and agency guidance, there is flexibility but also uncertainty about compliance. Overall, neither EU nor US approaches have fully resolved all AI-healthcare challenges (such as cross-border data sharing, or bias in diverse populations), but both models offer lessons for India.

The White House in 2022 issued a nonbinding "AI Bill of Rights" Blueprint, encouraging companies to conduct algorithmic impact assessments when deploying automated systems. The Blueprint is not law, but it reflects a consensus that AI risk must be documented and managed.

Many U.S. states are enacting privacy laws with AI provisions. Several (e.g. Colorado, California) require covered entities to perform data-protection impact assessments for profiling or automated decision making that affects consumers [[IAPP Article](#), Cole and Rolecki 2024, *Welcome to the 'Mild West': Balancing AI innovation, privacy principles*, 34]. Colorado even passed a specific AI law (effective 2026) defining "developers" and

“deployers” who must avoid discriminatory outcomes and can gain a legal defense by following recognized standards [[IAPP Article](#), Cole and Rolecki 2024, *Welcome to the 'Mild West:' Balancing AI innovation, privacy principles*, 34]. Enforcement agencies like the FTC also emphasize privacy-by design. In sum, U.S. policy now resembles a “mild West”: though fragmented, it emphasizes transparency, fairness audits, and adherence to privacy principles. Innovation is allowed to proceed, but companies are expected to embed privacy safeguards (e.g., privacy-by-design, data audits) into AI development.

This "vertical" regulatory model is composed of several key elements. First, sector-specific laws provide the primary legal guardrails. In healthcare, the Health Insurance Portability and Accountability Act (HIPAA) of 1996 governs the privacy and security of protected health information (PHI) [[IEEE Publication, 2023, What Is Digital Privacy and Its Importance?](#) 77]. While crucial, HIPAA was designed long before the advent of modern AI and does not address AI-specific challenges such as algorithmic bias or explainability.

Second, federal agencies are adapting their existing authorities to oversee AI within their respective domains. The Food and Drug Administration (FDA), for instance, has been actively developing a regulatory paradigm for "Software as a Medical Device" (SaMD), including AI-powered technologies. Recognizing that machine learning algorithms can evolve over time, the FDA is exploring frameworks that allow for predetermined change control plans, enabling manufacturers to update their models without requiring a full re-approval for every modification, provided the changes fall within a pre-specified scope. [[Lovells News Article 2](#), 2025, *AI Health Law & Policy: AI Health Law & Policy: FDA's rapidly evolving FDA's rapidly evolving regulatory paradigms*, 74] This approach is designed to be more adaptive to the dynamic nature of AI than traditional, static regulatory models. [[McNair et. al 2023](#), 7 *HEALTH CARE ARTIFICIAL INTELLIGENCE: LAW, REGULATION, AND POLICY*, 106]

Third, voluntary frameworks play a highly influential role in shaping industry best practices. The most prominent of these is the National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF). The AI RMF provides a structured, yet flexible, guide for organizations to manage AI risks throughout the system's lifecycle. It is organized around four core functions namely, Govern, Map, Measure, and Manage and promotes principles of trustworthiness, including validity, reliability, safety, security, transparency, explainability, privacy-enhancement, and fairness. While not legally binding, the NIST framework is widely seen as a benchmark for responsible AI development and is often referenced in policy discussions and corporate governance strategies.

3.3 Identifying Regulatory Tensions, Loopholes, and Key Challenges

The distinct approaches of the EU and the U.S. highlight a set of universal challenges that all regulatory systems for healthcare AI must confront. These challenges represent the core tensions at the intersection of technology, law, and ethics.

A primary challenge is **algorithmic bias**. AI systems trained on historical healthcare data can inherit and amplify existing societal biases, leading to discriminatory outcomes that worsen health disparities. The EU AI Act addresses this proactively by mandating high-quality, representative training data and bias-monitoring for high-risk systems. The U.S. approach is more reactive, relying on existing anti-discrimination laws and agency guidance to address bias after the fact. Case studies, such as algorithms that perform poorly in diagnosing skin cancer on darker skin tones or those that underestimate the health needs of minority patients, demonstrate the severe real-world consequences of this issue [Weber, 2023, [Hopkins Bloomberg Public Health Magazine](#), *Rooting Out AI's Biases*, 159]. Multiple sources warn of unfairness. [[Baumgartner et al. 2023](#), *Fair and equitable AI in biomedical research and healthcare: Social science perspectives*, 15] emphasize that AI can “perpetuate health inequalities for marginalized groups” if not carefully managed. Real-world examples include AI systems that inadvertently learn confounders or biases: Esmailzadeh cites neural nets “mis-diagnosing skin cancers by latching onto confounders like background colors” [[Esmailzadeh 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49]. Another example is telehealth expansion disproportionately benefited younger, non-minority patients [[Pesapane et al.](#)

[2021](#), *Legal and Regulatory Framework for AI Solutions in Healthcare in EU, US, China, and Russia: New Scenarios after a Pandemic*, 125], and AI trained on biased data can propagate inequities [[Leivaditis et al. 2025](#), *Artificial Intelligence in Thoracic Surgery: A Review Bridging Innovation and Clinical Practice for the Next Generation of Surgical Care*, 97]. Other reported cases involve risk algorithms that recommend lower levels of care for underserved patients or label minorities as high-risk disproportionately. Moreover, an AI trained on one population may not generalize to another; differences in demographics or care patterns mean a US-trained model may fail if deployed unchanged in Europe, and vice versa [[Haque et al. 2023](#), *Advancing Healthcare Outcomes with AI: Predicting Hospital Readmissions in the USA*, 67]. AI systems may also inherit human biases. The BMJ commentary cites a striking example: an AI suicide-risk calculator that had higher accuracy for White patients than Black patients, reflecting biased training data [[Ranjbari and Rahimi 2024](#), *Implications of conscious AI in primary healthcare*, 135]. Such outcomes would perpetuate inequity unless addressed by rebalancing data and algorithms. These biases can exacerbate disparities unless explicitly addressed. Explainability is cited as a related hurdle: without interpretable models, clinicians may distrust or misuse AI predictions. The suicide-risk example shows how biased AI can produce worse outcomes for minorities. Another scenario is patients receiving AI-generated information without context. The commentary describes a risk in palliative care: if an AI system predicts high mortality risk and automatically notifies a patient (or family) before a doctor can explain, it could “negatively impact the patient’s well-being” [[Ranjbari and Rahimi 2024](#), *Implications of conscious AI in primary healthcare*, 135]. This real-world pitfall arose from machine learning tools that helped clinicians know when to have end-of-life discussions, but without careful user interface design the tool might send raw predictions to patients.

Data security and privacy represent another critical hurdle. AI systems in healthcare process vast quantities of highly sensitive patient data, making them prime targets for cyberattacks. A data breach involving an AI system could expose the health records of millions, leading to a catastrophic loss of trust. While the GDPR provides a stringent and comprehensive standard for data protection, HIPAA's framework in the U.S. is often seen as less prescriptive, creating potential vulnerabilities that AI systems can exacerbate. AI depends on large, high-quality datasets. Poor data (missing, biased, or siloed) can degrade performance. Esmailzadeh’s survey enumerates “data scarcity” and fragmented records as critical issues [[Esmailzadeh 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49]. In practice, many AI studies are trained on narrow or retrospective cohorts; for example, one model validated on 175 cancer patients [[Alowais et al. 2023](#), *Revolutionizing healthcare: the role of artificial intelligence in clinical practice*, 8] may not generalize across diverse populations. Overcoming this requires extensive, representative training data – a challenge under current privacy constraints.

The **"black box" problem**, or the lack of transparency in complex AI models, poses a significant challenge to accountability and trust. [[The Regulatory Review](#), Cohen et al. 2022, *The AI Health Care Dilemma*, 33] It is often difficult, even for developers, to explain precisely why a deep learning model made a particular prediction or recommendation. This opacity conflicts with the medical and ethical need for clear, justifiable clinical reasoning. Regulators in both the EU and U.S. are pushing for greater "explainability," but this remains a technically challenging and sometimes elusive goal. The “black-box” nature of many deep learning models means clinicians and patients often cannot easily interpret AI outputs. In practice, informed consent forms rarely explain how an AI reaches its conclusion, nor the data it uses, undermining patient understanding. This lack of transparency can erode trust; [[Chau et al. 2025](#), *From black box to clarity: Strategies for effective AI informed consent in healthcare*, 27] note that current consent practices often “fall short of fully addressing the complexities” of AI, leaving patients unaware of algorithmic biases or data use. Both providers and patients may be uneasy with “black-box” AI. Ranjbari & Rahimi observe that patients often resist AI involvement in decisions, partly because “little is known about the exact steps through which AI connects an input to an output” [[Ranjbari and Rahimi 2024](#), *Implications of conscious AI in primary healthcare*, 135]. In shared decision making, lack of transparency can breed mistrust. Similarly, clinicians may be reluctant to act on AI alerts they do not understand. The authors suggest that fully explainable or even “conscious” AI might improve trust, but emphasize that until then, transparency and education are needed. Ethical concerns also arise in training and deployment: vast health-data sets fuel these models but raise privacy issues. For example,

microbiome profiles are intrinsically identifying and fall under strict EU GDPR rules, complicating data sharing. [Probul et al. 2024, *AI in microbiome-related healthcare*, 129] note that legal barriers and privacy regulations hinder pooling microbiome data internationally, a serious impediment given the heterogeneity needed for robust AI models. Clinicians are hesitant to rely on opaque AI recommendations [Pillay et al. 2025, *Artificial intelligence (AI) in point-of-care testing*, 126] [Choudhury and Asan 2022, *Impact of accountability, training, and human factors on the use of artificial intelligence in healthcare: Exploring the perceptions of healthcare practitioners in the US*, 29]. Explainable AI (XAI) helps: a German ICU case integrating LIME-based explanations saw clinician acceptance rise 40%. But XAI can trade off some accuracy – one trial found explainability reduced sensitivity by 7%. In nursing education, [Montejo et al. 2024, *Artificial intelligence (AI) applications in healthcare and considerations for nursing education*, 109] similarly warn that data security and algorithmic bias could exacerbate healthcare disparities if unaddressed; they call for new policies to govern AI development and deployment.

This leads directly to the question of **accountability and liability**. When an AI system contributes to a misdiagnosis or a harmful treatment decision, determining legal responsibility is complex. Is the developer liable for a flawed algorithm, the hospital for deploying the system, or the clinician for relying on its output? [WJBPHS Case Study, Chamarthy et al. 2024, *Artificial Intelligence in public health: A case study*, 25] The EU's framework attempts to create clearer lines of responsibility for providers and users of high-risk AI, while the U.S. system largely relies on existing tort and product liability law, which may be ill-suited for the dynamic nature of AI.

Explainability and oversight are key safeguards. The GDPR's transparency principle and the EU AI Act's documentation requirements both push for explainable systems. Ongoing monitoring is also crucial. In healthcare, regulators recommend continuous validation of AI tools (post-market surveillance) to catch biases that emerge over time. Similarly, educators must audit AI recommendations for fairness (e.g. checking that a tutoring AI does not systematically favor certain student groups). Training human stakeholders like teachers, doctors, students, patients about AI's capabilities and limits is an underutilized remedy. Empirical studies suggest that reminding users to maintain "vigilance" against AI errors (e.g. by requiring them to enter the final diagnosis or grade manually) can reduce automation bias [Goddard et al. 2011, *Automation bias: a systematic review of frequency, effect mediators, and mitigators*, 62].

Another challenge is **scope of rights**. Under GDPR, data subjects have rich rights (portable data, protest processing), but DPDP offers fewer. For example, GDPR allows a student or patient to ask for their data back and even to "port" it to another provider. DPDP also has a right to access and erasure, but no data portability clause. More critically, the lack of a "no automated decision" right in DPDP means AI driven admissions or diagnosis in India could proceed without a statutory opt-out. A possible solution is industry self-regulation: the EU AI Act and upcoming global best-practice guidelines (e.g. UNESCO's AI ethics, or professional standards in medicine/education) can fill normative gaps. Technically, developers can mitigate risks by design: using encrypted data storage, federated learning (so raw data never leave the institution), and privacy-preserving techniques. The Flint example shows one design fix: keeping student queries and responses internal to the school's environment [York School Study, Gao and Medved 2024, *How The York School used Flint to embrace student-centered AI use*, 58].

Finally, all regulatory systems grapple with the **pace of innovation**. AI technology is evolving at an exponential rate, and prescriptive, static regulations risk becoming obsolete quickly. The challenge of regulating adaptive algorithms that continuously learn and change "in the wild" versus "locked" algorithms that remain static after deployment is a key dilemma. The FDA's exploration of predetermined change control plans represents an attempt to create a more agile regulatory model, but this remains a frontier issue for regulators worldwide [EY Report, Dettling et al. 2022, *How the challenge of regulating AI in healthcare is escalating*, 42].

Chapter 4: AI in Practice: High-Impact Applications in US and EU Healthcare

Having established the foundational importance of robust legal frameworks and high-quality human capital, this analysis now examines the tangible outcomes of artificial intelligence deployment in developed healthcare systems. Through systematic review of high-impact applications and validated case studies from the United States and European Union, this chapter grounds the theoretical promise of AI in real-world evidence. The analysis demonstrates specific domains where AI delivers measurable improvements in efficiency, accuracy, and patient outcomes while identifying practical implementation challenges that inform evidence-based strategies for emerging healthcare markets.

Healthcare systems in the United States and Europe have begun deploying AI across multiple clinical domains with demonstrated success. In cardiology, neural networks measure ventricular function (ejection fraction) with exceptional accuracy (98%) and distinguish pathological from physiological hypertrophy in echocardiograms [Rudnicka et al., 2024, *Advancements in artificial intelligence-driven techniques for interventional cardiology*, 140]. AI triage tools, exemplified by the CEREBRIA-1 project, have matched expert cardiologists in planning coronary intervention. In pulmonary medicine, AI-enhanced imaging through radiomics and convolutional neural networks improves lung cancer detection and staging. Notably, an AI model for PD-L1 tumor scoring achieved 96.4% accuracy, matching expert pathologists [Leivaditis et al., 2025, *Artificial Intelligence in Thoracic Surgery: A Review Bridging Innovation and Clinical Practice for the Next Generation of Surgical Care*, 97]. In oncology radiology, AI segmentation of tumor volumes reduced contouring time by 43-56% without compromising accuracy.

4.1 Clinical Efficiency and Administrative Workflow Enhancement

Predictive analytics applications have demonstrated substantial impact on healthcare operations and patient outcomes. A compelling example emerges from Corewell Health in Michigan, where machine-learning models were deployed to identify patients at high risk of 30-day readmission [Haque et al., 2023, *Advancing Healthcare Outcomes with AI: Predicting Hospital Readmissions in the USA*, 67]. This AI system generated real-time alerts for clinicians, resulting in prevention of 200 readmissions over one year and estimated savings of \$5 million. The implementation produced "improved patient outcomes and quality of life" through timely interventions and "better resource utilization" as care teams could focus attention on the most vulnerable patients. This outcome demonstrates how predictive analytics can simultaneously improve clinical outcomes and reduce costs by enabling staff to allocate effort toward prevention rather than reactive care.

European implementations, while less extensively documented in explicit case reports, show similar applications in primary care settings. [Ranjibari and Rahimi, 2024, *Implications of conscious AI in primary healthcare*, 135] document AI utilization for screening and diagnosis tasks including early diabetic retinopathy detection from retinal images, atrial fibrillation risk prediction, and skin disease diagnosis through machine learning algorithms.

Administrative efficiency improvements have been realized through generative AI applications. Large language models can automate routine documentation by summarizing patient records, drafting discharge notes and progress reports, thereby alleviating physicians' clerical workload. As Reddy observes, generative AI "has the potential to transform healthcare through automated systems, enhanced clinical decision-making and democratization of expertise." These tools have been piloted in US hospitals and are under global evaluation to expedite documentation processes. AI chatbots have similarly been proposed for scheduling and claims processing, enabling patient interaction for appointment management based on clinician availability and automated insurance claims processing through form reading and verification. Such applications promise enhanced administrative efficiency through reduced time and errors while improving patient experience [Reddy, 2024, *Generative AI in healthcare: an implementation science informed translational path on application, integration and governance*, 137].

4.2 Medical Imaging and Diagnostic Revolution

AI applications in radiology and pathology have yielded marked improvements in detection accuracy across multiple specialties. In oncology imaging, [Chau et al., 2025, *From black box to clarity: Strategies for effective AI informed consent in healthcare*, 27] report that an AI system trained on mammograms from 25,856 UK and 3,097 US women reduced both false-positive and false-negative rates, achieving higher accuracy than human radiologists. Machine-learning algorithms now automatically detect lesions on X-rays and CT scans, accelerating diagnoses and reducing missed findings. AI tools for radiology quality control flag suboptimal images to ensure only high-quality scans inform clinical decisions while optimizing imaging protocols to minimize patient radiation exposure [Dave and Patel, 2023, *Artificial intelligence in healthcare and education*, 37].

Pathology applications have demonstrated human-level performance in critical diagnostic tasks. In the CAMELYON16 challenge, an ensemble of 32 convolutional neural networks achieved an area under the curve (AUC) of 0.99 for detecting breast cancer in lymph node histology, essentially matching expert pathologists. Similar AI image-analysis models segment tumors and predict cancer prognosis, with specialized tools analyzing chromatin images to flag early breast cancer changes and fusing ultrasound with blood-marker data through LightGBM models to predict metastasis with higher AUC than standard clinical models, [Chau et al., 2025, *From black box to clarity: Strategies for effective AI informed consent in healthcare*, 27].

Advanced "augmented intelligence" systems exemplify the integration of AI into complex clinical decision-making. DeepCoro in cardiology ingests entire angiography videos to quantify coronary stenoses, having been trained on approximately 182,000 studies to achieve an AUROC of 0.829 (accuracy 79%) in estimating narrowing severity with lower inter-observer variability than cardiologists. By automating feature extraction from live angiograms, this tool enables more consistent revascularization decisions than visual inspection [Langlais et al., 2024, *DeepCoro: Automated Coronary Stenosis Quantification from Angiography*, 94].

Deep-learning models have significantly outperformed traditional methods in cancer detection across multiple organ systems. One convolutional neural network achieved breast cancer detection sensitivity of 50.9% versus 22.4% using conventional breast-density measures. Similar systems for pancreatic and lung cancer have reported accuracies of 80-95%. In cardiac care, AI-enhanced electrocardiogram analysis improved heart-failure prediction AUC from 0.756 to 0.818 by combining ECG data with clinical risk factors. These results demonstrate AI's capacity for improved early detection, with [Khosravan and Bagci, 2018, *Deep Learning for Lung Nodule Detection*, 87]'s deep-learning algorithm identifying lung nodules at 95% accuracy compared to 65% for radiologists [Manickam et al., 2022, *Artificial Intelligence (AI) and Internet of Medical Things (IoMT) Assisted Biomedical Systems for Intelligent Healthcare*, 100].

4.3 Treatment Personalization and Prognostic Enhancement

Clinical decision support systems have demonstrated robust applications in personalizing therapy through patient-specific data analysis. [Huang et al., 2018, *Machine Learning for Chemotherapy Response Prediction*, 75] trained a support-vector machine learning algorithm on gene expression data from 175 cancer patients to predict chemotherapy response with greater than 80% accuracy. Similarly, AI models utilizing electronic health record data from approximately 17,500 patients predicted individual responses to different antidepressants with high reliability.

Precision dosing represents a particularly successful application domain. In a multicenter study, an AI model trained on warfarin patients (19,719 inpatients) "outperformed expert physicians" in predicting future INR values, yielding more reliable individualized dosing [Alowais et al., 2023, *AI in Medical Education: Applications and Future Directions*, 8]. The CURATE.AI platform dynamically optimized chemotherapy doses in a prospective trial for advanced solid tumors, with CURATE.AI integration showing "successful incorporation and potential benefits" by reducing dosage while improving tumor response and duration compared to standard care. These validated improvements in patient management and safety demonstrate AI's capacity to enhance treatment planning and dosing precision.

4.4 Clinical Decision Support and Professional Education

AI-driven clinical decision support systems are being integrated into hospital information technology workflows, though implementation challenges persist. [[Hoffmann et al., 2024](#), *SMART-on-FHIR Platform for Ophthalmology AI Integration*, 72] describe a SMART-on-FHIR platform for ophthalmology AI that addresses administrative hurdles by using interoperability standards (FHIR) to link AI tools with electronic health records. However, they note that "AI-based CDSS...are rarely used in clinical practice, partly due to lack of interoperability with clinical information systems (CIS)." The proposed platform enables launching AI-CDSS from anywhere in a hospital system given FHIR compatibility.

Hospitals in both the European Union and United States face similar implementation challenges including fragmented electronic health records and proprietary data formats that hinder AI deployment. Successful AI-CDSS integration, particularly for FDA-approved imaging software, requires alignment with regulatory standards such as the EU's Medical Device Regulation or US FDA requirements [[Esmailzadeh, 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49], along with standardized data formats to enable interoperability.

In nursing and clinical support applications, AI focuses on education and decision support enhancement. Clinical decision support systems integrated into electronic records generate patient-specific alerts and order sets, including automated warnings for drug allergies, evidence-based order set suggestions for given diagnoses, and reminders for overdue preventive care [[Montejo et al., 2024](#), *Artificial intelligence (AI) applications in healthcare and considerations for nursing education*, 109]. These AI-augmented systems have demonstrated improved workflow efficiency and reduced medication errors, though careful calibration is required to avoid alert fatigue.

Educational applications show promise in professional development. Virtual patient simulations and chatbots are emerging as training tools, with one pilot utilizing a conversational "virtual patient" chatbot to educate heart failure patients remotely, improving engagement and adherence. In nursing education, Shorey et al. trained a virtual counseling chatbot for student role-playing, with nursing learners showing higher self-efficacy and confidence in clinical communication after chatbot interaction [[Montejo et al., 2024](#), *Artificial intelligence (AI) applications in healthcare and considerations for nursing education*, 109]. AI-powered simulators enable dental and medical students to practice procedures in virtual reality, such as performing fillings or root canal preparations on digital patients before treating actual patients. These systems can reduce educator workloads while standardizing training through automated grading and feedback tools.

Dental applications demonstrate AI's expanding scope in healthcare education and practice. AI algorithms analyze radiographic dental images to detect cavities or periodontal disease earlier than unaided visual examination. Virtual reality simulators allow dental students to rehearse complex procedures including root canals and crown preparations on digital patients. AI can enhance patient engagement through smart toothbrushes and applications that monitor oral hygiene and recommend personalized care routines. In medical-dental education, AI-generated examination questions and automated peer-feedback systems are being tested to provide objective student assessments and reduce instructor bias [[Dave and Patel, 2023](#), *Artificial intelligence in healthcare and education*, 37].

4.5 Telemedicine and Remote Monitoring Expansion

AI-powered Internet-of-Medical-Things (IoMT) devices have significantly expanded point-of-care capabilities and remote monitoring options. On-demand telehealth systems such as TytoPro enable remote physical examinations under AI guidance. Wearable monitors and implanted sensors, combined with AI analytics, allow continuous tracking of vital signs including ECG monitoring for atrial fibrillation and various physiological markers. These systems have demonstrated capacity to "reduce unnecessary hospital stays and associated health costs" by enabling early intervention outside hospital settings.

Chronic disease management benefits substantially from miniaturized IoMT devices. Glucose monitors that feed data into AI algorithms for real-time alerts exemplify this application, with AI-based glucose sensing

systems being prototyped for non-invasive monitoring [[Manickam et al., 2022](#), *Artificial Intelligence (AI) and Internet of Medical Things (IoMT) Assisted Biomedical Systems for Intelligent Healthcare*, 100].

The COVID-19 pandemic accelerated telehealth adoption through regulatory changes and AI-driven platform enhancements. US telemedicine usage surged when Medicare allowed full reimbursement, with UC San Diego Health expanding from approximately 870 tele-visits over three years to 119,500 visits in five months during 2020. European countries similarly relaxed regulations, enabling 70% higher usage of services such as the UK's "Push Doctor" platform [[Pesapane et al., 2021](#), *Legal and Regulatory Framework for AI Solutions in Healthcare in EU, US, China, and Russia: New Scenarios after a Pandemic*, 125]. Virtual care platforms utilize AI for triage and monitoring, with RNN-based wearable analytics such as DeepHeart using Fitbit data to predict cardiovascular risk better than standard biomarkers [[Rudnicka et al., 2024](#), *Advancements in artificial intelligence-driven techniques for interventional cardiology*, 140]

Point-of-care testing and primary care have incorporated AI to enhance on-site diagnostic capabilities. AI-powered portable devices significantly improve diagnostic accuracy, with deep-learning analysis of blood smears detecting malaria parasites with 95% sensitivity, vastly superior to unassisted microscopy. Additional point-of-care examples include AI-based anaemia screening with approximately 94% accuracy and smart glucose monitors that reduce faulty readings by roughly 30% [[Pillay et al., 2025](#), *AI-Powered Point-of-Care Diagnostic Devices*, 126]. Predictive analytics maintain device uptime with 20% reduction in downtime, enabling faster results.

In primary care settings, FDA-cleared AI systems such as IDx-DR autonomously screen for diabetic retinopathy, substantially improving early intervention for diabetes-related vision loss [[Liyanage et al., 2019](#), *Artificial Intelligence in Primary Health Care: Perceptions, Issues, and Challenges*, 98]. AI models for sepsis prediction deployed in emergency settings have achieved high accuracy (AUC ~0.89) [[Pillay et al., 2025](#), *AI-Powered Point-of-Care Diagnostic Devices*, 126], enabling earlier lifesaving alerts. Overall, AI in these settings enhances both diagnostic accuracy through pattern recognition and workflow efficiency through automated routine interpretations.

4.6 Specialty Care Applications and Professional Development

AI applications in specialized medical fields demonstrate expanding capabilities across diverse clinical domains. In neurology, AI-enhanced neuroimaging supports early Alzheimer's diagnosis through deep learning models that analyze MRI/PET scans and genetic markers to detect subtle biomarker patterns. Natural language processing tools analyze speech and cognitive tests to identify early dementia signs [[Kale et al., 2024](#), *AI-driven innovations in Alzheimer's disease: Integrating early diagnosis, personalized treatment, and prognostic modelling*, 84].

Andrology applications exemplify AI's precision in specialized diagnostics. Convolutional neural networks have classified sperm images into WHO morphology categories with approximately 94% true-positive rates. AI-powered Computer-Assisted Semen Analysis (CASA) systems analyze sperm motility and capacitation with enhanced accuracy, while artificial neural network models predict fertility potential from lifestyle questionnaires. A smartphone-based home sperm test using AI image analysis accurately measures motility and concentration, offering potential for decentralized fertility screening [[Ghayda et al., 2024](#), *Artificial Intelligence in Andrology: From Semen Analysis to Image Diagnostics*, 60].

These specialized applications yield higher diagnostic precision and enable patient-tailored insights. As noted in comprehensive reviews, AI in Alzheimer's and other specialized areas "can improve early diagnosis, personalize treatments, and predict disease outcomes more accurately," ultimately enhancing quality of life [[Esmaeilzadeh, 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49].

Primary care implementation research reveals both potential and current limitations. A systematic review found that 45% of community-based primary care AI studies utilized machine learning for predictive analytics while

27% employed natural language processing. Most implementations targeted diagnosis, detection, or surveillance tasks such as identifying diabetes or hypertension. AI-enabled tools have been tested for patient triage, routine examination support, and chronic disease monitoring. However, the same review highlights that very few AI systems have achieved effective deployment at scale in community settings, observing "a large gap in the effective development and implementation of AI in community-based primary health care" [[Rahimi et al., 2021](#), *Application of Artificial Intelligence in Community-Based Primary Health Care: Systematic Scoping Review and Critical Appraisal*, 132]. This finding suggests that most primary care AI applications remain experimental, pending evidence of real-world benefit.

4.7 Regulatory Framework Challenges and Implementation Barriers

Regulatory oversight of healthcare AI varies significantly between jurisdictions, creating compliance complexities that impede widespread adoption. In the United States, the FDA has begun approving AI-enabled devices, primarily imaging software, under existing medical device legislation [[Esmailzadeh, 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49]. The Federal Trade Commission additionally oversees bias and claims in AI products. However, specialists argue that current regulations including FDA and HIPAA frameworks only partially address AI's unique challenges.

Recent US policy initiatives attempt to fill regulatory gaps. The "AI Bill of Rights" (2022) and Executive Order (2023) establish principles for safe, transparent, and equitable AI deployment. In the European Union, legislators have pursued more comprehensive approaches through the EU AI Act (agreed December 2023), which represents the first comprehensive AI regulation. This legislation categorizes medical AI as high-risk and imposes strict requirements including transparency, human oversight, and bias testing [[Esmailzadeh, 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49].

Pre-pandemic restrictive regulations significantly hampered telemedicine adoption. US state licensure rules prevented out-of-state remote care, with 49 states requiring in-state licensing [[Pesapane et al., 2021](#), *Legal and Regulatory Framework for AI Solutions in Healthcare in EU, US, China, and Russia: New Scenarios after a Pandemic*, 125], while Medicare provided minimal reimbursement for tele-visits. Only emergency waivers including Medicare parity and relaxed HIPAA enforcement enabled widespread adoption. In diagnostics, unclear regulatory pathways delayed AI implementation, with FDA clearance for AI mammography requiring years of advocacy.

Europe's Medical Device Regulation (active May 2021) explicitly covers AI-based software as medical devices, requiring harmonized risk-based reviews [[Pesapane et al., 2021](#), *Legal and Regulatory Framework for AI Solutions in Healthcare in EU, US, China, and Russia: New Scenarios after a Pandemic*, 125]. However, this regulatory transition temporarily delayed some approvals as manufacturers adapted to new requirements. Data privacy laws further complicate AI deployment, with the EU's GDPR (2018) imposing strict consent, breach notification, and explainability mandates [[Pillay et al., 2025](#), *AI-Powered Point-of-Care Diagnostic Devices*, 126], while US HIPAA provides weaker, sector-specific protections.

Multiple reviews emphasize that healthcare AI development often outpaces legislative frameworks. [[Chau et al., 2025](#), *From black box to clarity: Strategies for effective AI informed consent in healthcare*, 27] explicitly note that "many jurisdictions lack comprehensive regulations" tailored to AI's unique challenges. This regulatory gap leads to inconsistent implementation across regions, with tools approved in one jurisdiction potentially forbidden or unregulated in another, creating complications for multi-center trials and deployment scaling. An AI tool developed for US hospitals may require extensive revalidation under EU medical device rules before European use. Similarly, no standardized GDPR-compliant framework exists for global AI datasets, potentially preventing validated diagnostic tools from reaching international clinics.

4.7.1 Implementation Failures and Systemic Barriers

While explicit system failures are rarely documented in detail, several sources indicate significant caution regarding deployment risks. Rahimi et al. note "doubts concerning the unintended harmful effects" of deploying AI in primary care without evidence-based oversight [[Rahimi et al., 2021](#), *Application of Artificial Intelligence in Community-Based Primary Health Care: Systematic Scoping Review and Critical Appraisal*, 132]. [[Esmailzadeh, 2024](#), *Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations*, 49] identify examples where AI could be perceived as threatening, including job displacement and loss of clinical autonomy, if not implemented thoughtfully. Notably, severe patient harm cases are not cited in reviewed sources, suggesting either underreporting or that many systems remain in trial phases. However, authors uniformly emphasize that potential harms including diagnostic error amplification and privacy breaches must be mitigated through robust design and review processes.

Despite documented successes, literature consistently identifies barriers that limit AI's current impact. Fragmented data infrastructure represents a fundamental challenge, with US hospitals and health systems typically creating isolated data "silos" within proprietary EMR systems and local networks. Vendor-locked EMR systems and lack of interoperability significantly complicate aggregation of the large, diverse datasets required for AI model training and validation. Algorithms developed on one hospital's data may not generalize to other settings, with AI trained on predominantly Caucasian populations "not expected to have the same accuracy when applied to minorities" [[Panch et al., 2019](#), *The "inconvenient truth" about AI in healthcare*, 121]. Poor EMR usability further degrades data quality through inconsistent clinical record entry, while interoperability "remains elusive." Data fragmentation and quality issues including incomplete, noisy, or biased records therefore constrain AI performance [[Haque et al., 2023](#), *Advancing Healthcare Outcomes with AI: Predicting Hospital Readmissions in the USA*, 67]

4.7.2 Workflow Integration and Human Factors

Even high-performing AI models provide limited value without successful integration into daily clinical practice. Retrospective deployment reviews identify that the primary challenge involves "integrating this into established workflow" and training staff on new processes [[Haque et al., 2023](#), *Advancing Healthcare Outcomes with AI: Predicting Hospital Readmissions in the USA*, 67]. Clinician resistance to unfamiliar tools requires cultural change and extensive training to prevent adoption stagnation. Healthcare organizations often lack in-house expertise or capital to build and maintain advanced AI infrastructure including cloud computing and high-performance data systems, placing them behind other industries in technological adoption.

Technology alone cannot transform healthcare delivery. AI systems must align with institutional incentives, data practices, and clinician workflows, yet many proposed solutions have failed to "re-engineer the incentives" of fragmented healthcare systems [[Panch et al., 2019](#), *The "inconvenient truth" about AI in healthcare*, 121]. Successful AI implementation requires comprehensive organizational change beyond technological deployment, encompassing workflow redesign, staff training, and institutional culture adaptation.

4.8 Flo Health Class Action Lawsuit Over Health Data Sharing

Flo Health's Flo Period & Ovulation Tracker, a leading app for women's fertility and menstrual health, became the target of privacy litigation after reports that it was transmitting users' highly sensitive health data to third-party advertisers. Investigators found that Flo had incorporated Meta (Facebook) and Google analytics tools into the app in ways that exposed individual users' menstrual and reproductive details, contrary to the company's earlier privacy promises [[Federal Trade Commission](#) Report, 2021, *FTC Finalizes Order with Flo Health, a Fertility-Tracking App that Shared Sensitive Health Data with Facebook, Google, and Others*, 56] [[Reuters Report](#), Green, 2025, *As class action trial looms, Meta and Flo could face 'mind-boggling' damages*, 65]. In January 2021 the U.S. Federal Trade Commission (FTC) charged that Flo misled consumers by promising confidentiality while sharing private health information with analytics firms, and Flo settled that case in June 2021 under a consent order requiring affirmative user consent and notice whenever health data are disclosed. Soon thereafter several class actions were filed in California (e.g. Gamino v. Flo and Frasco v. Flo, 2021)

alleging that Flo's app by integrating Google's and Meta's software development kits – automatically collected and transmitted intimate user information (period dates, pregnancy intent, sexual activity, etc.) to third parties without proper notice or consent [[Courthouse News Service](#), Attridge, 2025, *Trial begins in data privacy case against menstrual tracking app Flo*, 13].

The plaintiffs assert violations of California's Confidentiality of Medical Information Act and Invasion of Privacy Act (CMIA and CIPA, which impose statutory damages of roughly \$1,000–\$5,000 per violation) and other privacy laws, and they claim Flo breached its privacy promises by effectively “selling” or otherwise misusing deeply personal reproductive data for targeted advertising [[Bloomberg Law](#), Poritz, 2025, *Period Tracking App Users Win Class Status in Google, Meta Suit*, 128]. By contrast, Flo (represented by Dechert LLP) and Meta deny any illegal conduct. They emphasize that Flo's privacy policy did disclose certain analytics uses and that any data transmitted were de-identified (device-code hashes) and used only for app maintenance or improvement.

Upon downloading Flo Health's period and ovulation tracking application, users provide personal identifiers like name, age, location alongside deeply sensitive reproductive data such as last menstrual dates, fertility intentions, and menopausal status. The disclosures ignited widespread consumer backlash after a 2019 Wall Street Journal exposé, prompting the U.S. Federal Trade Commission to charge Flo with deceptive practices and issue a 2021 settlement mandating affirmative consent before any health data sharing [[Reuters Report](#), Green, 2025, *As class action trial looms, Meta and Flo could face 'mind-boggling' damages*, 65]. In the ensuing litigation, analytics firm Flurry agreed in March 2025 to a \$3.5 million settlement to resolve claims of unauthorized harvesting of Flo users' health information [[ClassAction.org](#), Mehorter 2025, *\$3.5 Million Flo App Settlement Aims to Resolve Data Harvesting Class Action Claims Against Flurry*, 108], while Google reached an undisclosed accord, leaving Flo and Meta to defend against class certification and liability in a July 2025 trial [[Reuters Report](#), Green, 2025, *As class action trial looms, Meta and Flo could face 'mind-boggling' damages*, 65]. These legal actions coincide with Flo's reported \$200 million revenue in 2023, underscoring the app's profitability amid a free-to-user model monetized via data-driven advertising [[Business of Apps](#), Curry 2025, *Flo Revenue and Usage Statistics (2025)*, 21].

Flo maintains it did not “sell” personal data, arguing that any shared information was de-identified, disclosed in its privacy policy, and covered by user agreements, whereas plaintiffs contend that statutory violations of the California Confidentiality of Medical Information Act and Invasion of Privacy Act entitle affected users to significant damages. This consolidated litigation highlights the tension between digital health innovation and regulatory expectations for transparent, consent-based handling of sensitive health data.

In court filings and opening statements Flo's attorneys have stressed that the company never “sold” users' personal data and that users affirmatively agreed to the relevant terms (including a one-year limitations clause), noting that the app expressly disclaims any status as a medical provider [[Courthouse News Service](#), Attridge, 2025, *Trial begins in data privacy case against menstrual tracking app Flo*, 13]. Meta's counsel similarly argues that Facebook did not intentionally eavesdrop on the data and that developers agree not to send any protected health information through Facebook's tools [[Reuters Report](#), Green, 2025, *As class action trial looms, Meta and Flo could face 'mind-boggling' damages*, 65]

In May 2025 Judge James Donato of the Northern District of California certified both nationwide and California subclasses, finding common questions about disclosure and consent; as he wrote, a user's “loss of control over one's personal information” can itself be a concrete harm.

By mid-2025, three of Flo's co-defendants have resolved their exposure: analytics firm Flurry agreed to pay \$3.5 million in a March 2025 settlement, Google reached an undisclosed settlement around July 2025, and claims against AppsFlyer were dismissed. Flo and Meta are proceeding to a jury trial in July 2025 on the remaining claims. No class verdict has been reached yet. The litigation underscores regulators' and courts' focus on health-app privacy: it comes on the heels of the FTC order and it illustrates how traditional privacy statutes and contract promises are being applied to modern health-tracking platforms [[Federal Trade](#)

[Commission](#) Report, 2021, *FTC Finalizes Order with Flo Health, a Fertility-Tracking App that Shared Sensitive Health Data with Facebook, Google, and Others*, 56] [[Reuters Report](#), Green, 2025, *As class action trial looms, Meta and Flo could face 'mind-boggling' damages*, 65].

An important thread is how humans misunderstand or misuse AI in these fields. In healthcare, automation bias and “black box” explanations are major issues. Studies show clinicians sometimes place too much trust in AI recommendations: one systematic review found that while clinical decision-support systems generally improve accuracy, users often fail to notice the new types of errors AI can introduce. Automation bias leads to “errors of commission” (accepting wrong AI advice) and “omission” (overlooking patient details because AI did not flag them) [[Goddard et al. 2011](#), *Automation bias: a systematic review of frequency, effect mediators, and mitigators*, 62]. In cognitive terms, when under stress or workload, doctors may see AI as an “advice” dispenser and defer to it. Conversely, if the AI seems uncertain, doctors might ignore valuable suggestions.

From the user side, patients may also be confused about AI’s role. For example, a patient might not realize whether an online health chatbot is providing human medical advice or purely algorithmic guidance, raising consent issues. Overall, these human–AI mismatches point to the need for explainability and oversight. Regulatory frameworks begin to address this: GDPR’s Article 22 ensures a human remains in the loop for significant decisions, and the EU AI Act requires documentation so decisions can be audited. Ethically, institutions are encouraged to educate staff on AI limitations and to involve users in design. In healthcare, it means clinicians must retain final judgment and patients should consent to AI-assisted care.

Hospital AI systems are subject to stringent consent and security rules. Patients must be informed if their data are used beyond treatment (e.g. in research or by third-party providers). Breach notifications are mandatory: if an AI system leaks health data, regulators expect fines up to 4% of turnover. The European Health Data Space (EHDS) initiative also promotes secondary use of health data under GDPR-compliant conditions, enabling AI research with safeguards.

These challenges underscore the trade-off between the EU’s “horizontal” approach, which offers consistency at the risk of rigidity, and the U.S.’s “vertical” model, which provides flexibility at the cost of fragmentation. The optimal path for other nations, including India, may lie in a hybrid approach that combines the strengths of both: a strong, horizontal foundation for core principles like data protection, complemented by agile, vertical regulations for high-risk sectors like healthcare.

The evidence from US and EU healthcare systems demonstrates that AI applications can deliver substantial improvements in clinical efficiency, diagnostic accuracy, and patient outcomes across multiple medical domains. From predictive analytics preventing hospital readmissions to AI-enhanced imaging achieving superior diagnostic performance, validated implementations provide compelling evidence for AI’s transformative potential in healthcare delivery.

However, successful deployment requires careful attention to regulatory frameworks, data infrastructure, and workflow integration challenges. The gap between AI’s proven capabilities in controlled settings and effective real-world implementation highlights the importance of comprehensive strategies that address technical, organizational, and regulatory dimensions simultaneously. For emerging healthcare markets, these lessons emphasize the critical importance of building robust foundational infrastructure alongside AI capability development to realize the technology’s full potential in improving patient care and system efficiency.

Chapter 5: Machine Learning and LLMs in US and EU Healthcare: Applications, Impact, and Challenges

5.1 Machine Learning Models in Healthcare Settings

Machine learning (ML) techniques have been increasingly integrated into healthcare in both the United States and the European Union over the past two decades. Hundreds of ML-enabled medical devices and software have been approved: for example, by 2020 there were an estimated 222 AI/ML-based medical devices authorized in the US and 240 in Europe [[Muehlematter et al. 2021](#), *Approval of artificial intelligence and machine learning-based medical devices in the USA and Europe (2015-20): a comparative analysis*, 111]. These approvals have grown rapidly since 2015, especially in imaging and diagnostics. Notably, many devices common to both regions were first cleared in Europe (80 of 124 such devices), suggesting earlier or less stringent evaluation in some European pathways. However, neither region originally had a dedicated regulatory framework for adaptive AI/ML systems. This regulatory gap is now being addressed (e.g. the EU's new AI Act of 2024 requires risk mitigation and oversight for "high-risk" medical AI [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52]), but its absence has led to both innovation and pitfalls.

5.1.1 Diagnostic and Screening Applications

Imaging and Radiology - ML is widely used to interpret medical images. In radiology and pathology, deep-learning algorithms have been trained to detect anomalies on X-rays, CT/MRI scans, and digital slides. For instance, an AI system developed by Google/DeepMind with Moorfields Eye Hospital (London) was trained on thousands of retinal OCT scans and achieved diagnostic accuracy comparable to world-leading ophthalmologists. In published trials it matched experts' referral decisions for over 50 eye diseases with about 94% accuracy [[Moorfields Eye Hospital \(NHS\)](#), 2019, 110]. Such AI could help screen for diabetic retinopathy (diagnostic imaging now uses AI support tools where there is now FDA-approved algorithms for diabetic retinopathy screening are used in US primary care [[Liyanage et al. 2019](#), *Artificial Intelligence in Primary Health Care: Perceptions, Issues, and Challenges*, 98]), age-related macular degeneration, and other sight-threatening conditions earlier than routine review.

Breast cancer screening has also seen notable ML applications. In a recent large German study, AI was integrated into the national mammography screening program as an aid to radiologists. Compared to standard double-reading, AI-assisted double-reading increased the cancer detection rate by ~17.6% (from 5.7 to 6.7 per 1,000 women) without increasing the recall rate [[Eisemann et al. 2025](#), *Nationwide real-world implementation of AI for cancer detection in population-based mammography screening*, 47]. Other prospective trials worldwide (e.g. South Korea's AI-STREAM) have similarly shown that AI support can raise cancer detection rates while maintaining or even reducing false-positive recalls. These results suggest ML can improve early diagnosis in population screening, potentially improving outcomes (at the cost of additional work-ups).

In acute care, ML is used to triage imaging in real time. For example, FDA-cleared software like Viz.ai's stroke-detection tool automatically scans head CT angiograms for large-vessel occlusions (LVOs). Deployments in >1,700 hospitals have demonstrated dramatic workflow benefits: one multicenter study reported AI-assisted care shortened time-to-treatment (door-to-puncture) by ~52 minutes and reduced length of stay by about 2.5 days [[Viz.ai](#), 156]. Viz.ai's system alerts stroke teams to suspected occlusions within seconds of imaging and ensures specialists see 90% of alerts within 5 minutes. Overall, evidence suggests ML can streamline time-critical imaging triage and lead to faster interventions and better functional outcomes in stroke care.

Predictive Analytics and Clinical Decision Support - Beyond imaging, ML is applied to electronic health record (EHR) data and bedside monitoring to predict patient risks. One well-studied example is sepsis prediction. Researchers at MIT and Massachusetts General Hospital developed an ML model to guide early sepsis therapy decisions. Trained on emergency-department sepsis patient data, the model predicted within

hours which patients would require vasopressor medications with over 80% accuracy [[MIT News](#), 2018, Matheson, *Machine-learning system could aid critical decisions in sepsis care*, 102]. In practice, such a model (if implemented as a real-time alert) could prompt earlier treatment changes that improve survival. Similarly, many hospitals use ML-based “early warning” tools to predict ICU transfers, readmissions, or cardiac arrest risk by learning patterns in vital signs and lab values. (For example, modified versions of the “EWS” score now often incorporate ML). There is also active use of ML in drug discovery and clinical trials: algorithms scan biomedical literature and molecular libraries to suggest novel drug candidates or optimize trial design (an area being fostered by regulators).

Administrative and Operational Tasks - ML is adopted for non-clinical efficiency as well. In hospital operations, predictive models forecast patient admissions, optimize bed and staffing allocation, and manage patient flow. The EU Commission notes that “predictive modelling can forecast patient admissions and optimise the use of hospital beds, staff, and equipment” [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52]. Hospitals in the US similarly experiment with ML to minimize wait times or predict high-volume periods. ML also automates administrative work: for instance, it can auto-populate EHR forms, sort medical coding, or streamline billing. On the patient side, ML-driven chatbots (e.g. symptom checkers) have been piloted to triage non-urgent queries, aiming to reduce unnecessary clinic visits (though their performance is variable).

5.1.2 IBM Watson Oncology

AI faces pitfalls in healthcare, was illustrated by IBM’s Watson for Oncology. Watson was marketed as a decision support system to help oncologists recommend cancer treatments, but it encountered serious safety and transparency problems. In practice, clinicians found that Watson sometimes gave “unsafe and incorrect” treatment recommendations. These errors raised immediate concerns about patient safety and the reliability of AI-driven advice in clinical settings.

The root cause of Watson’s problems turned out to be its training data. Watson’s oncology model had been trained primarily on “synthetic” cancer patient cases authored by doctors, instead of on large datasets of real patient records. Because those contrived cases did not capture the full variability of real cancers, Watson learned from misleading patterns. This failure exemplifies the “garbage in, garbage out” principle: unreliable or incomplete training data led the AI to produce unreliable recommendations. In other words, without real-world data to ground its learning, Watson’s outputs were prone to serious mistakes.

Internal documents revealed that it sometimes suggested chemotherapy combined with Bevacizumab for bleeding patients, a lethal combination that “shouldn’t be given” due to hemorrhage risk [[AIAAIC Repository](#), 2018, *IBM Watson recommends “unsafe and incorrect” cancer treatments*, 3]. These errors (unfortunately kept secret for over a year) prompted clinicians to question the system’s reliability. Although no patient was harmed (the unsafe advice was not implemented), the incident highlights that ML models can produce dangerous recommendations if trained on inappropriate data. It underscored the need for clinical validation and transparency: “the revelations raised serious questions about the reliability and safety of AI in healthcare” [[AIAAIC Repository](#), 2018, *IBM Watson recommends “unsafe and incorrect” cancer treatments*, 3].

Another significant issue was the lack of transparency from IBM about Watson’s shortcomings. Reports indicate that IBM kept the fact of Watson’s “unsafe and incorrect” recommendations secret for more than a year. This decision obscured an important problem from clinicians and patients and undermined confidence in the technology. The Watson case highlights the necessity of openness: AI developers must be transparent about the data used to train their systems and forthcoming about any known limitations or errors. Only by disclosing these details can clinicians and patients trust that they understand how the AI arrived at its advice.

5.1.3 Failures in Practice: The Challenge of Accountability and Liability

Many early ML tools were introduced before clear rules existed, leading to trust and legal issues. In 2017, the UK's Information Commissioner ruled that the Royal Free London NHS Trust had unlawfully shared the medical records of 1.6 million patients with Google DeepMind for a pilot app (Streams for kidney injury) without proper consent and notification [[The Guardian](#), Hern 2017, *Royal Free breached UK data law in 1.6m patient deal with Google's DeepMind*, 70]. Patients “would not have reasonably expected” their data to be used for such testing, the ICO found. This high-profile case demonstrated that insufficient legal compliance can derail AI projects and erode public trust. It also illustrated the need for transparent legal frameworks: the ICO noted the Data Protection Act is “not a barrier to innovation, but it does need to be considered wherever people’s data is being used” [[The Guardian](#), Hern 2017, *Royal Free breached UK data law in 1.6m patient deal with Google's DeepMind*, 70].

Accountability and Liability - It is often unclear who is responsible when an ML tool errs. For example, with autonomous screening AI in ophthalmology, liability typically falls on the device manufacturer for missed findings [[Retina Specialist](#), Anand et al. 2025, *AI for DR screening: Where are we in 2025?*, 11], since the software (not the clinician) made the decision. However, ambiguity arises when multiple conditions are visible in one image; the screen-and-churn device might miss a serious non-target condition (e.g. retinal detachment). The referenced retinal article points out that if the AI misses such pathology, it is “less clear who’s responsible”. Similarly, in cardiology devices or clinical decision support, it is debated whether doctors or hospitals bear the fault for following a flawed algorithm. This legal uncertainty can impede adoption: as regulators note, robust liability and compensation rules are needed so that “victims can effectively seek compensation from manufacturers” in case of AI-related harm [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52].

5.1.4 The Regulatory Response: EU and US Frameworks for AI in Healthcare

In recent years, regulators have begun to catch up. In the EU, the 2024 AI Act explicitly classifies AI intended for medical purposes as “high-risk.” Such systems must meet stringent requirements: risk management plans, high-quality training data, human oversight, and transparency provisions [[European Commission Report - Artificial Intelligence in healthcare](#), 2022, 52]. The Act’s enactment reflects recognition that ML in healthcare can significantly affect patient welfare, and robust safeguards are needed. The EU also emphasizes patient rights: the Commission stresses the need for frameworks allowing victims to seek redress for AI-caused harm.

In the US, the FDA has been developing policies for AI/ML medical devices. The agency has published guidance on “Software as a Medical Device” and proposed frameworks (e.g. the 2019 discussion paper on Predetermined Change Control Plans) to handle continuously learning algorithms [[FDA Discussion Paper, 2019](#), *Proposed Regulatory Framework for Modifications to Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) - Discussion Paper and Request for Feedback*, 57]. Current practice generally requires pre-market review of substantial software changes. However, as a Lancet analysis notes, the FDA and EU had no dedicated regulatory pathway for AI/ML devices until very recently [[Muehlematter et al. 2021](#), *Approval of artificial intelligence and machine learning-based medical devices in the USA and Europe (2015-20): a comparative analysis*, 111]. This historical gap meant many ML tools were evaluated as conventional software, with variable transparency. Experts have called for more open registries of approved algorithms and clarity on oversight to build public trust. In practice, FDA-cleared ML devices are still relatively few, and most are categorized as lower-risk (Class II) with 510(k) clearance.

Additionally, existing health privacy laws constrain ML development. In the EU, the General Data Protection Regulation (GDPR) imposes strict rules on patient data use (including data minimization and patient consent), and even grants individuals a right not to be subject to solely automated decisions affecting them. In the US, HIPAA governs protected health information, and ML developers must ensure compliance. The Royal Free case [[The Guardian](#), Hern 2017, *Royal Free breached UK data law in 1.6m patient deal with Google's DeepMind*, 70] and similar incidents underscore that strong legal frameworks are needed: without them, well-intentioned ML projects can run afoul of privacy and ethics standards.

5.2 The Rise of Large Language Models (LLMs) in Clinical Workflows

Large language models (LLMs) have begun appearing in actual healthcare services worldwide, often in the form of chatbots or decision-support tools. For example, [Xiao et al. 2023, *Powering an AI Chatbot with Expert Sourcing to Support Credible Health Information Access*, 163] deployed a COVID-19 “Jennifer” chatbot on Facebook (and other sites) that draws on expert sources to answer users’ queries. They found that Jennifer helped users get more accurate COVID-related information. In Japan, researchers have even used LLMs on Twitter data to gauge public sentiment during the pandemic, demonstrating useful health-insight mining [Sarker et al. 2025, *Natural Language Processing for Digital Health in the Era of Large Language Models*, 142]. Similarly, patient-facing bots like Woebot (a text-based mental health coach) and the UK’s SERMO chatbot have been adopted to support behavioral health. In a clinical trial, Woebot – powered by conversational AI which significantly reduced depressive symptoms compared to controls [Yang et al. 2023, *Large language models in health care: Development, applications, and challenges*, 166], showing real benefit from LLM-driven therapy. By automating low-level tasks, clinicians have more time for patients. For example, large language models (LLMs) embedded in electronic records can digest complex patient data and produce concise summaries or draft notes, saving clinicians hours each day. One review highlights that AI documentation tools could “alleviate the current documentation burden and the consequent burnout many physicians” face [Reddy 2024, *Generative AI in healthcare: an implementation science informed translational path on application, integration and governance*, 137]. These examples illustrate LLMs serving on the patient side: answering health questions on social media and providing cognitive/ behavioral support in everyday care settings [Yang et al. 2023, *Large language models in health care: Development, applications, and challenges*, 166].

5.2.1 ChatGPT in Healthcare as LLM

Clinical documentation, communication, and Electronic Health Record Integration - Several studies have explored using LLMs to generate medical text with minimal clinician input. For example, Lancet Digital Health reported that ChatGPT could draft patient clinic letters (for skin cancer care) with high quality: independent reviewers gave GPT-generated letters median scores of 7/10 for factual correctness and “humanness” [Ali et al. 2023, *Using ChatGPT to write patient clinic letters*, 7]. These letters were written at the recommended 11–12 year reading level, improving patient comprehension. The authors noted that AI could save clinician time and enhance clarity, but stressed that outputs must be carefully reviewed by physicians to catch omissions or errors. Similarly, a 2025 BMC Health Services study found that GPT-3.5 ChatGPT could synthesize long medical notes into concise inpatient discharge summaries. In hypothetical cases from an infectious-disease service, ChatGPT produced well-structured discharge reports summarizing hospital course and follow-up plans [Koh et al. 2025, *Using ChatGPT for writing hospital inpatient discharge summaries – perspectives from an inpatient infectious diseases service*, 90]. Authors concluded ChatGPT “provided an effective framework” for these summaries, though “close reading” was needed to verify accuracy. In critical care, an Irish pilot study showed GPT-4 generated readable ICU discharge summaries from free-text notes. GPT-4 captured about 80% of key events (better than GPT-3.5 and Llama2) but still missed some details and introduced narrative errors [Urquhart et al. 2024, *A pilot feasibility study comparing large language models in extracting key information from ICU patient text records from an Irish population*, 155]. These examples illustrate that LLMs can accelerate documentation by summarizing complex records or drafting patient letters, but that human oversight remains essential to prevent misstatements.

Patient-facing communication - In one experiment, ChatGPT was tasked with rewriting clinic letters into lay language for patients. It maintained factual accuracy in most cases and produced readable text, suggesting LLMs could help simplify clinical information for patients (though formal evaluation of comprehension was not reported) [Ali et al. 2023, *Using ChatGPT to write patient clinic letters*, 7].

Beyond chatbots, LLMs have been deployed as **patient education and counseling aids**. For instance, an animated avatar called “Macy” was built using ChatGPT to deliver medication counseling via video. Macy conveyed dosage instructions, side-effect warnings, and red-flag symptoms in plain language – all generated by the LLM [Yang et al. 2023, *Large language models in health care: Development, applications, and challenges*, 166].

In an experimental trial, Macy was produced in under 30 minutes at virtually no cost, highlighting how LLMs can rapidly personalize patient education. Similarly, the virtual respiratory clinic AnsibleHealth used ChatGPT to translate technical radiology reports and medical records into patient-friendly summaries. These real-world pilots (though still early-stage) demonstrate that LLMs can bridge communication gaps, boosting health literacy by turning medical jargon into accessible explanations for patients [[Yang et al. 2023](#), *Large language models in health care: Development, applications, and challenges*, 166].

5.2.2 Applications in Clinical Practice and Workflow

Electronic Health Record (EHR) - The integration of LLMs into Electronic Health Record (EHR) systems represents one of the most significant developments in healthcare technology. Epic, the largest EHR vendor in the United States with 35.9% market share, has partnered with Microsoft to integrate Azure OpenAI Service into its platform [[FIERCE Healthcare Article](#), Landi, 2024, *HIMSS24: How Epic is building out AI, ambient technology for clinicians*, 93]. Through this collaboration, Epic has developed over 60 use cases for LLM integration, ranging from automated patient message responses to clinical note generation. The implementation has yielded measurable results across multiple health systems. Epic reported that over 150,000 notes have been drafted using ambient technology, with multiple millions of In Basket message draft responses created. At Mayo Clinic, initial pilots showed that AI-enhanced messaging saved nurses approximately 30 seconds per message and generated more empathetic responses. The health system plans to expand access to all licensed practical nurses and registered nurses, potentially saving 1,500 hours per month. Stanford Health Care, UC San Diego Health, and UW Health were among the first institutions to deploy Epic's LLM-integrated solutions for automatically drafting responses to patient messages. The integration also includes Nuance's DAX Copilot, which uses ambient AI technology to automate clinical documentation by "listening" to physician-patient conversations and generating draft clinical notes within seconds [[FIERCE Healthcare Article](#), Landi, 2024, *HIMSS24: How Epic is building out AI, ambient technology for clinicians*, 93]. Valley Medical Center implemented the CORTEX AI solution for utilization management, achieving a 67% improvement in case review volume and enabling completion of all reviews compared to just 60% prior to implementation [[SciMedian Article](#), Pathak, 10 *Case Studies of Successful Implementation of AI in Healthcare By SciMedian*, 124]. This allowed nurses to focus on clinical merits rather than solely on criteria-based assessments.

Google's DeepMind has achieved remarkable success in ophthalmology, developing an AI algorithm that can detect over 50 eye diseases from retinal scans with high accuracy [[SciMedian Article](#), Pathak, 10 *Case Studies of Successful Implementation of AI in Healthcare By SciMedian*, 124]. This advancement has facilitated early intervention and improved patient outcomes, addressing the critical challenge of timely disease recognition in ophthalmology practice.

LLMs are also entering clinical workflow and documentation. For example, hospital teams have experimented with ChatGPT/GPT-4 to generate discharge summaries. [[Patel and Lam 2023](#), *ChatGPT: the future of discharge summaries?* 123] reported that ChatGPT could quickly produce draft discharge letters from physician notes, potentially reducing doctors' paperwork. In radiology, one study found that ChatGPT could correctly recommend the next imaging steps for breast cancer screening and evaluation of breast pain – suggesting LLMs might aid diagnostic decision support [[Sallam 2023](#), *The Utility of ChatGPT as an Example of Large Language Models in Healthcare Education, Research and Practice: Systematic Review on the Future Perspectives and Potential Limitations*, 141]. More broadly, LLM-powered tools have been trialed to summarize patient EHR histories, extract key clinical data, translate records into billing codes, and even automate routine responses to patients (e.g. appointment reminders) [[Yang et al. 2023](#), *Large language models in health care: Development, applications, and challenges*, 166]. These deployments have improved efficiency in tested settings, though they also disrupt workflows: clinicians must now review and verify AI-generated notes and alerts.

Clinical decision support - LLMs have also been evaluated for diagnostic and triage assistance, often with mixed results. A multi-center German study compared GPT-4-based ChatGPT to emergency physicians on ED triage cases. Remarkably, GPT-4 achieved substantial agreement with expert consensus (weighted $\kappa \approx 0.67$), on par with novice doctors [[Masanneck et al. 2024](#), *Triage Performance Across Large Language Models, ChatGPT, and*

Untrained Doctors in Emergency Medicine: Comparative Study, 101]. It significantly outperformed GPT-3.5 ($\kappa \approx 0.54$). However, even GPT-4 did not reach the level of senior clinicians, and using ChatGPT as a second opinion did not significantly improve triage accuracy over human judgement. The authors cautioned that while LLMs may match inexperienced clinicians in raw triage proficiency, they are not yet “gold-standard” decision-makers [[Masannek et al. 2024](#), *Triage Performance Across Large Language Models, ChatGPT, and Untrained Doctors in Emergency Medicine: Comparative Study, 101*].

Likewise, a randomized clinical trial at two U.S. academic centers tested whether ChatGPT could boost physicians’ diagnostic reasoning. Physicians (attendings and residents) were given case vignettes either with or without ChatGPT access. Median diagnostic accuracy was ~76% with ChatGPT versus 74% without (a non-significant difference). The study concluded that simply providing an LLM “did not improve overall physician diagnostic reasoning” [[Goh et al. 2024](#), *Large Language Model Influence on Diagnostic Reasoning, 63*]. In fact, treating GPT alone (with no human) would have solved 92% of cases, but in practice doctors using it scored similarly to controls. The investigators warned that LLMs should not be used autonomously for diagnosis and must remain under physician oversight.

5.2.3 Administrative and Research Applications

Clinical Trial Optimization and Research Support - LLMs have demonstrated significant potential in streamlining clinical trial processes. A novel framework utilizing LLMs for cancer clinical trials showed accuracy rates peaking at over 92% with datasets consisting of thousands of patient data points. The system demonstrated 87% accuracy for SNOMED-CT, 90% for LOINC, and 84% for RxNorm codes, significantly outperforming existing models like GPT-4 and Claude 3.5, which peaked at an average of 77% [[Shekhar and Kim 2024](#), *Novel Development of LLM Driven mCODE Data Model for Improved Clinical Trial Matching to Enable Standardization and Interoperability in Oncology Research, 146*]. Research has shown that LLM-based patient-trial matching can improve performance by 7.32% on average, with generalizability to new data improved by 12.12% [[Yuan et al. 2023](#), *Large Language Models for Healthcare Data Augmentation: An Example on Patient-Trial Matching, 167*]. This addresses significant challenges in clinical trial enrollment, where traditional methods are often manual, time consuming, and lack a data-driven approach. Novo Nordisk has reported significant success with their Novoscribe system, which reduced the time to produce clinical study reports from 90 days to one week and decreased the number of people needed for this process from 42 to just three [[Waracle Article](#), 2024, *Revolutionising Clinical Trials with Large Language Models, 158*]. This demonstrates the potential for LLMs to dramatically improve efficiency in clinical research documentation and reporting.

Administrative Efficiency and Cost Reduction - The implementation of LLMs in administrative tasks has yielded significant efficiency gains. DeepScribe's integration with Epic EHR systems has enabled some healthcare providers to reduce after-hours EHR documentation by up to 75%. The system's proprietary HEAL LLM delivers accuracy rates 32% better than platforms running GPT-4 as a standalone model [[Informa TechTarget Article](#), Nelson 2024, *AI Medical Scribe Deepens Epic EHR Integration to Enhance Workflows, 120*]. TidalHealth Peninsula Regional achieved substantial improvements in clinical decision support by implementing IBM Micromedex with Watson, reducing time per clinical search from 3-4 minutes to less than one minute [[Xsolis Article](#), 164]. This translates to significant time savings for healthcare providers, allowing more time for direct patient care. A large urban hospital network that adopted AI-driven predictive analytics for readmission prevention achieved a 20% reduction in 30-day readmission rates and realized annual cost savings of \$4 million while improving patient satisfaction scores [[Medwave LLC](#), 2024, *How AI is Transforming Healthcare: 12 Real-World Use Cases, 107*].

Data and administrative automation - LLMs have shown promise in processing unstructured data. For instance, GPT-4 was used to automatically de-identify oncology notes: it could identify and redact protected health information with high precision and recall. This enabled generation of “synthetic” clinical notes that preserve patient privacy. The study’s authors noted this could “revolutionize” medical data use by expanding available research datasets without compromising privacy [[Altalla et al. 2025](#), *Evaluating GPT models for clinical note de-identification, 9*]. In administration, AI-enabled “scribes” are emerging. NHS England recently released

guidance to safely adopt AI-powered ambient scribing (voice-to-text) systems in hospitals and clinics. These tools capture spoken consultations and generate draft notes or even letters. The guidance emphasizes that outputs must be reviewed by clinicians and comply with all data-protection and medical-device standards [National Health Service, UK, 2025, *Guidance on the use of AI-enabled ambient scribing products in health and care settings*, 118], reflecting the regulatory scrutiny around new LLM-based documentation tools.

Research and Development Breakthroughs - Several institutions have achieved significant advances in medical research through LLM implementation. Atomwise has utilized its AtomNet platform to accelerate drug discovery, successfully identifying potential treatments for the Ebola virus within a single day [SciMedian Article, Pathak, *10 Case Studies of Successful Implementation of AI in Healthcare By SciMedian*, 124]. This represents a dramatic reduction in drug discovery timelines from years to months, demonstrating the transformative potential of LLMs in pharmaceutical research.

The Mayo Clinic's collaboration with Google Cloud has resulted in an AI and machine learning platform that aids in patient care and research, including algorithms that automatically perform complex calculations such as assessing breast cancer risk. This platform provides clinicians with diverse tools for improved patient outcomes while supporting ongoing research initiatives.

Boston Children's Hospital's implementation of AI-powered virtual nursing assistants has enhanced patient care quality while improving workflow efficiency [SciMedian Article, Pathak, *10 Case Studies of Successful Implementation of AI in Healthcare By SciMedian*, 124]. These voice-activated assistants, integrated with hospital systems, allow nurses to manage workloads more effectively, resulting in improved patient interaction and overall care quality.

5.2.4 Significant Failures, Risks, and Limitations

Ethical failings have also surfaced. Hallucinations - confident but false statements remain a problem, especially in generative outputs. As Reddy cautions, powerful LLMs “demonstrate impressive fluency,” but when used in “high-stakes” domains like healthcare they can confidently produce entirely fabricated information [Reddy 2024, *Generative AI in healthcare: an implementation science informed translational path on application, integration and governance*, 137]. A hallucinated diagnosis or treatment plan could mislead clinicians and directly harm patients if unchecked. For instance, an LLM might invent a benign explanation for a serious symptom; if a clinician were unaware of the error, the patient’s condition could worsen. No concrete clinical failure of this sort is documented in these sources, but the risk is repeatedly noted as a failure mode. Clinicians found that ChatGPT sometimes has undergone these hallucinations, generating plausible but false medical advice or fabricated citations. Two case reports in a journal-writing contest showed ChatGPT-produced articles containing nonexistent references and factual errors, leading editors to strongly caution against unsupervised use. Although no publicly documented patient harm has been reported yet, experts warn this is dangerous: “generation of inaccurate content can have severe negative consequences in healthcare” [Sallam 2023, *The Utility of ChatGPT as an Example of Large Language Models in Healthcare Education, Research and Practice: Systematic Review on the Future Perspectives and Potential Limitations*, 141]. Likewise, tests revealed that GPT-4 can exhibit bias in medical judgments across gender, race or ethnicity [Sarker et al. 2025, *Natural Language Processing for Digital Health in the Era of Large Language Models*, 142]. Such biases prompted calls for legal controls on LLMs in medicine. The net effect is that, in practice, every AI-generated suggestion or document must be carefully checked by humans. Where LLM outputs improve efficiency, they also force workflow changes: providers now must act as gatekeepers, reviewing AI summaries, counterchecking LLM recommendations, and educating patients about AI limitations. One report noted that GPT-3.5 would have suggested a “fatal” treatment for intestinal obstruction, a critical error that GPT-4 corrected. More broadly, LLMs inherit biases from training data [Clusmann et al. 2023, *The future landscape of large language models in medicine*, 32] and lack explainability, raising concerns for high-stakes decisions. In the psychiatric note study (unpublished), five categories of clinically relevant errors were found when converting clinician notes into patient-friendly language: misinterpretations (e.g. suicidality), attributions, distorted content, incorrect abbreviations, and syntax errors [Ali et al. 2023, *Using ChatGPT to write patient clinic letters*, 7]. These errors

could “impair clinical judgment” and patient understanding, underscoring the need for caution and human oversight.

Additionally, LLM performance varies by task: GPT-4 beat GPT-3.5 and smaller models on most tasks in studies [[Urquhart et al. 2024](#), *A pilot feasibility study comparing large language models in extracting key information from ICU patient text records from an Irish population*, 155]. Thus, early deployments relying on less-capable LLMs may underperform. In the triage study, all LLMs tended to overtriage patients (erring on the side of caution), whereas doctors sometimes undertriaged [[Masanneck et al. 2024](#), *Triage Performance Across Large Language Models, ChatGPT, and Untrained Doctors in Emergency Medicine: Comparative Study*, 101]. This misalignment suggests further tuning or hybrid workflows are needed.

Finally, concrete failures have surfaced. In practice, healthcare workers have occasionally uploaded real patient PHI into ChatGPT and similar services without realizing the privacy risk. A recent survey found 81% of data-policy violations in hospitals involved sensitive health data being shared with cloud or AI apps. This has led to warnings that using non - HIPAA compliant AI tools with patient data can violate privacy laws [[The HIPAA Journal Report](#), Alder 2025, *Healthcare Workers Violating Patient Privacy by Uploading Sensitive Data to GenAI and Cloud Accounts*, 6]. In one US case, a nurse pasted a patient’s identifiable notes into ChatGPT and OpenAI later notified the hospital under their HIPAA policy. These incidents highlight that misuses of LLMs can quickly raise legal and trust issues unless tightly controlled.

5.2.5 Regulatory Landscape and Compliance Challenges

United States FDA Approach

The Food and Drug Administration has established a comprehensive framework for regulating AI/ML-enabled medical devices. As of August 2024, the FDA has authorized 950 AI or machine learning-enabled medical devices, with 221 devices authorized in 2023 alone [[Encord Blog](#), 2024, *The Step-by-Step Guide to Getting Your AI Models Through FDA Approval*, 48][[MedTech Dive Article](#), Reuter and Han, 2024, *The number of AI medical devices has spiked in the past decade*, 139]. This represents a dramatic increase from just six authorized devices in 2015, highlighting the accelerating adoption of AI in healthcare.

The FDA primarily uses three pathways for AI medical device approval: 510(k) clearance, Premarket Approval (PMA), and De Novo classification. Remarkably, 96.7% of approved AI/ML enabled devices have been cleared through the 510(k) pathway, which allows for marketing devices that are "substantially equivalent" to already legally marketed devices. Only 0.4% (3 devices) have received approval via the more stringent PMA pathway [[Ailotte Article](#), 2025, *FDA AI Medical Device Compliance: What Your Business Must Know Now*, 4].

The FDA's approach emphasizes several key principles: ensuring AI technologies provide real value to patients, maintaining the "human-in-the-loop" principle for oversight, and requiring transparency in how AI systems function [[IQVIA Article](#), King 2024, *The Future of AI in Medical Devices: FDA Guidelines and Global Perspectives*, 80]. In December 2024, the FDA finalized recommendations for streamlining the approval process through Predetermined Change Control Plans (PCCPs), which allow manufacturers to modify AI systems without requiring additional marketing submissions for each change [[American Hospital Association](#), 2024, *FDA finalizes recommendations simplifying approval process for medical devices that use AI*, 10]

However, significant challenges remain in FDA regulation of LLMs specifically. The agency has not yet authorized any technology that uses generative AI or LLMs as medical devices, primarily because most approved medical devices have "locked algorithms" rather than continuously learning systems [[Healthcare Information and Management Systems Society](#), Harding et al. 2024, *Section 5 - Navigating the Regulatory Landscape: An Analysis of Legal and Ethical Oversight for Large Language Models (LLMs)*, 68]. The FDA faces the challenge of regulating adaptive algorithms and autodidactic learning systems that can modify their behavior over time.

Cross-Border Regulatory Challenges

The divergent approaches between the FDA and EU AI Act create significant challenges for organizations developing healthcare AI solutions for global markets. While the FDA focuses on device-specific approvals through established pathways, the EU AI Act takes a broader horizontal approach that applies across all AI applications in healthcare.

The EU's approach emphasizes risk-based classification and requires comprehensive documentation of AI system development, training data, and decision-making processes. In contrast, the FDA's framework, while requiring similar safety and effectiveness demonstrations, operates within the established medical device regulatory structure with emphasis on predicate device comparisons and clinical validation.

Healthcare organizations operating internationally must navigate these different regulatory frameworks simultaneously. The lack of harmonization between US and EU approaches creates additional compliance burdens and may slow the deployment of beneficial AI technologies in healthcare settings.

Privacy and Data Protection Violations

The integration of LLMs into healthcare systems has raised significant privacy and data protection concerns. Patient data used to interact with LLMs may be collected and analyzed, potentially violating HIPAA regulations and other privacy protections. The European GDPR presents additional challenges, as LLMs may process personal health information in ways that are not compliant with strict European data protection requirements.

Research has shown that patient data can potentially be reidentified through cross-referencing with other databases, even when supposedly deidentified [[Kim and Vajravelu 2025](#), *Assessing the Current Limitations of Large Language Models in Advancing Health Care Education*, 88]. The integration of LLMs with electronic health records and medical record systems greatly heightens privacy concerns, as it involves exchange of information between multiple databases that may reduce the effectiveness of data anonymization efforts.

Legal, Regulatory Considerations and Ethical Challenges

Real-world LLM use in healthcare has immediately highlighted legal and governance gaps. For instance, because closed-source LLMs like ChatGPT process data via off-site servers, hospitals discovered that patient information could accidentally be transmitted externally. Lacking clear laws, some healthcare systems imposed internal bans or guidelines on ChatGPT use, while others negotiated “secure transfer” agreements with AI vendors. Experts stress that institutions should obtain informed consent before using LLM tools on patient data [[Sarker et al. 2025](#), *Natural Language Processing for Digital Health in the Era of Large Language Models*, 142]. In the EU and US, new regulations are emerging: the EU’s AI Act and the U.S. HIPAA now require rigorous documentation, data encryption, and risk assessments for AI in healthcare [[Nazi and Peng 2024](#), *Large Language Models in Healthcare and Medical Domain: A Review*, 119]. However, many real deployments preceded such rules. A recent review noted that some LLM systems were actually put into use “without regulatory oversight and without human control of quality, truthfulness or appropriateness” [[Harrer 2023](#), *Attention is not all you need: the complicated case of ethically using large language models in healthcare and medicine*, 69] – a situation that leaves patients and providers exposed. In the absence of clear liability rules, any LLM-driven medical error could provoke litigation: commentators warn that issues of medico-legal accountability for AI mistakes (e.g. misdiagnosis from an LLM) remain unsettled [[Sallam 2023](#), *The Utility of ChatGPT as an Example of Large Language Models in Healthcare Education, Research and Practice: Systematic Review on the Future Perspectives and Potential Limitations*, 141]. In short, hospitals had to improvise privacy and safety controls in the regulatory vacuum.

Neither the EU nor the US currently has AI-specific healthcare laws, so existing regulations (privacy, medical devices, etc.) apply to LLM use. In the US, HIPAA governs patient data. ChatGPT is not HIPAA compliant out of the box, so using it on identifiable health records violates the rule. As noted, many clinicians have inadvertently breached HIPAA by using ChatGPT for documentation [[The HIPAA Journal Report](#), Alder 2025, *Healthcare Workers Violating Patient Privacy by Uploading Sensitive Data to GenAI and Cloud Accounts*, 6]. In the EU, the GDPR severely restricts processing of personal health data. A European ICU study pointed out that GDPR

requires individual consent for processing clinical notes, which “prohibits mass processing of individual patient data” for LLM training. This regulatory environment has hindered large-scale LLM research in Europe: only a handful of EU studies exist, compared to many US projects that rely on deidentified or public data [[Urquhart et al. 2024](#), *A pilot feasibility study comparing large language models in extracting key information from ICU patient text records from an Irish population*, 155]. The authors argue that new “legislative changes” are needed so that researchers can ethically access health records for AI development.

Looking ahead, the forthcoming EU AI Act will likely classify AI systems used for medical triage or diagnosis as “high risk”, imposing strict oversight (data quality, documentation, human control). NHS guidance already requires medical AI scribes to meet relevant safety standards (England’s DCB0160) and medical-device rules [[National Health Service, UK](#), 2025, *Guidance on the use of AI-enabled ambient scribing products in health and care settings*, 118]. In the US, FDA oversight is evolving: software that “interprets or analyzes” patient data may be regulated as a medical device, but draft guidance is still broad. Liability also remains unresolved. Legal experts warn that without clear rules on AI errors, it is uncertain who is responsible if an LLM-guided decision harms a patient. One commentary notes that “clear rules and legal boundaries are needed” to allocate liability and protect users [[Wang et al. 2023](#), *Ethical Considerations of Using ChatGPT in Health Care*, 157].

Outcomes and Outlook

In practice, the benefits of deployed LLMs are tangible but nuanced. Patient education bots and automated note-writing tools have saved time and broadened access to medical information [[Yang et al. 2023](#), *Large language models in health care: Development, applications, and challenges*, 166] [[Sallam 2023](#), *The Utility of ChatGPT as an Example of Large Language Models in Healthcare Education, Research and Practice: Systematic Review on the Future Perspectives and Potential Limitations*, 141]. For example, ChatGPT-generated discharge letters have markedly reduced physicians’ documentation burden in pilot projects. On the positive side, patients have received more understandable explanations (from bots like Macy and AnsibleHealth’s system), and researchers have sped up tasks like protocol drafting. Yet every benefit has come with new costs: clinicians must navigate untested AI advice, technical staff must audit data use, and institutions must absorb the burden of creating AI policies without a clear playbook.

Notably, the literature documents no single “AI scandal” or patient tragedy yet attributable to LLM use. However, multiple papers emphasize that fundamental safeguards were lacking in these deployments. In one commentary authors call for “clear legal frameworks for the responsible use of LLMs in healthcare”, noting that without them every use case – from patient chatbots to draft reports – carries unanswered legal and ethical questions [[Nazi and Peng 2024](#), *Large Language Models in Healthcare and Medical Domain: A Review*, 119] [[Sarker et al. 2025](#), *Natural Language Processing for Digital Health in the Era of Large Language Models*, 142]. Until such frameworks exist, healthcare providers worldwide will continue to experience both the rapid gains and the disruptive side-effects of LLMs in practice.

In summary, while leading hospitals and agencies are beginning to issue guidance, many LLM uses remain in a gray zone. Real world examples underscore the stakes: inappropriate LLM use has already led to privacy breaches and patient-safety alerts. Both EU and US authorities are now scrambling to fill these gaps. Pending specific laws, best practice is cautious: any LLM output should be carefully verified by clinicians, and training/deployment must comply with existing health data regulations [[Wang et al. 2023](#), *Ethical Considerations of Using ChatGPT in Health Care*, 157] [[The HIPAA Journal Report](#), Alder 2025, *Healthcare Workers Violating Patient Privacy by Uploading Sensitive Data to GenAI and Cloud Accounts*, 6].

Chapter 6: Comparative Perspective US vs EU: Adoption, Opportunities, and Regulatory Context

Both US and EU healthcare see similar potential from AI, but their approaches and contexts differ. In the US, innovation often comes from hospitals and startups implementing AI for concrete financial and clinical gains. The Corewell case shows a focus on cost savings under Medicare/Medicaid and quality metrics [[Haque et al.](#)

[2023](#), *Advancing Healthcare Outcomes with AI: Predicting Hospital Readmissions in the USA*, 67]. Regulatory attention tends to use existing laws (HIPAA, FDA reviews) rather than new AI-specific rules. In the EU, application is often guided by policy: the forthcoming EU AI Act exemplifies a “risk-based” stance. European health systems are cautious about privacy (GDPR) and equity. For instance, the referenced BMJ commentary (from Europe/Canada) frames AI use in terms of ethics and human values [[Ranjbari and Rahimi 2024](#), *Implications of conscious AI in primary healthcare*, 135]. In practice, U.S. health systems have aggressively pursued AI - driven efficiency and clinical gains, whereas European systems approach AI through a precautionary, policy led lens emphasizing data protection and equity.

6.1 AI Adoption in U.S. Healthcare

U.S. healthcare is largely a mix of private and public systems; innovation often emerges from hospitals and health networks seeking financial and clinical returns. Regulatory oversight focuses on patient data privacy (HIPAA) and medical device safety (FDA). HIPAA mandates safeguards on “protected health information” but permits data use for treatment and operations, allowing many AI pilots to proceed under existing rules [[Censinet Article](#), *GDPR vs. HIPAA: Key Differences for Healthcare*, 24]. The FDA reviews AI as software or medical devices, recently issuing “Good Machine Learning Practices” for model transparency and oversight [[The Institute for Experiential AI, Northeastern University](#), Trovato 2025, *The Myths We Still Believe About AI in Healthcare*, 153].

Recent U.S. surveys document rapid AI uptake in healthcare. A 2025 nationwide survey of 43 U.S. health systems (64% response rate) found widespread deployment of AI tools. For example, Ambient Clinical Notes (a generative AI transcription tool) was deployed or piloted in 100% of surveyed systems, with 53% of respondents reporting high success in clinical documentation tasks. AI use in radiology and imaging was also prevalent: 90% of systems had at least partial AI use in imaging workflows (e.g. chest x-ray triage or cancer scans), although only a minority reported high diagnostic accuracy gains. Predictive analytics (such as early sepsis alerts or readmission risk scores) were commonly adopted, though only ~38% of hospitals rated those tools as highly effective [[Poon et al. 2025](#), *Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges*, 127].

Physician attitudes in the U.S. underscore practical gains. A 2025 AMA survey found many doctors enthusiastic about AI, especially for administrative automation [[Medscape Medical News](#), McKenna 2025, *Doctors Warm to AI Future in Healthcare, AMA Survey Finds*, 104]. Most respondents noted AI’s biggest impact in streamlining clerical work: automated transcription of notes, scheduling, and patient messaging were cited as primary AI uses. This aligns with industry reports: one survey found U.S. hospitals using AI first to automate revenue-cycle and staffing tasks, yielding concrete financial improvements (e.g. a 30% revenue lift from AI-driven patient communications). In short, U.S. adoption is hospital-driven and ROI focused, with providers piloting AI to cut costs, reduce burnout, and improve care workflows.

Nonetheless, U.S. systems face barriers. The JAMIA survey highlights that immature AI tools (identified by 77% of respondents) and upfront costs (47%) are major obstacles [[Poon et al. 2025](#), *Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges*, 127]. Regulatory uncertainty also rated highly (40% cited it), reflecting questions about liability when AI errs. Still, U.S. institutions generally navigate HIPAA (which applies only to covered entities and allows broad data de-identification) and rely on FDA review for clinical tools. As one analysis notes, U.S. governance emphasizes “human oversight” and adaptability: the FDA’s guidelines stress clinician-in-the-loop transparency [[The Institute for Experiential AI, Northeastern University](#), Trovato 2025, *The Myths We Still Believe About AI in Healthcare*, 153], contrasting with Europe’s more prescriptive approach.

In summary, U.S. AI in healthcare is characterized by rapid, bottom-up innovation within hospitals and health systems. Examples include AI-assisted radiology (e.g. FDA-cleared X-ray and CT algorithms), clinical support tools (e.g. Epic’s sepsis alerts or diabetes retinopathy screening), and workflow automation (e.g. AI patient-call systems). These applications are pursued for demonstrable financial and clinical outcomes, and the evolving

U.S. policy (HIPAA, FDA guidance) generally allows such innovation to proceed, provided patient data are protected.

6.2 AI Adoption in EU Healthcare

Europe's healthcare landscape is a patchwork of public systems, but the EU aims for harmonized AI strategy. European policy emphasizes "the EU way": trustworthy, ethical AI aligned with human rights [[McKinsey and Company Report](#), 2020, *Transforming healthcare with AI*, 105]. Importantly, the GDPR (2018) imposes stringent data protection: all personal health data are "special category" requiring explicit consent or strict safeguards [[Censinet Article](#), *GDPR vs. HIPAA: Key Differences for Healthcare*, 24]. Unlike HIPAA, GDPR applies extraterritorially – any U.S. tech firm processing EU health data must comply. This privacy-first environment shapes how hospitals use AI.

European research suggests AI is mostly at pilot stage. A 2020 EIT Health/McKinsey survey found that only 44% of EU healthcare organizations had even developed or deployed an AI application [[McKinsey and Company Report](#), 2020, *Transforming healthcare with AI*, 105]. When AI is used, the focus is often on diagnostics and decision-support within hospitals. For example, a study of nine European hospitals identified 11 AI use-cases spanning diagnosis, patient care, and logistics [[Klumpp et al. 2021](#), *Artificial Intelligence for Hospital Health Care: Application Cases and Answers to Challenges in European Hospitals*, 89]. These included AI-driven coronary artery disease diagnosis, automated arrhythmia detection, fetal monitoring analysis, and even rare disease pattern recognition. In care management, use-cases involved chronic disease management engines, resource allocation tools, and adverse-event prediction. In logistics, hospitals piloted AI for material consumption forecasting and robotic assistance in wards. This diversity illustrates Europe's broad interest: AI could aid everything from imaging interpretation to optimizing bed assignments [[Klumpp et al. 2021](#), *Artificial Intelligence for Hospital Health Care: Application Cases and Answers to Challenges in European Hospitals*, 89].

However, European deployments are generally cautious and evaluated. The same hospital case-study report notes that expected AI benefits are often tempered by concerns about data privacy, consent, and patient acceptance. Echoing this, the McKinsey report observes that despite "significant opportunity", "AI's full potential...remains to be explored and the impact on the ground remains small" [[McKinsey and Company Report](#), 2020, *Transforming healthcare with AI*, 105]. Part of the lag reflects fragmented data infrastructures: even though EU systems hold a "treasure trove" of medical records, linking and sharing data across regions is difficult under strict GDPR rules.

Policy is a major driver in Europe. The EU AI Act (effective August 2024) enshrines a risk-based approach: virtually all clinical AI (e.g. diagnostic algorithms, personalized medicine tools) is classified "high-risk" [[Aboy et al. 2024](#), *Navigating the EU AI Act: implications for regulated digital medical products*, 1]. High-risk AI in healthcare must meet heavy compliance (rigorous risk management, transparency, quality systems, etc.). For instance, developers must document training data, bias assessment, and post market surveillance. This dovetails with existing laws: AI-driven devices already fall under the EU Medical Device Regulation (MDR) and must also satisfy GDPR's data-use constraints. In practice, the EU Act and GDPR together mean AI projects undergo thorough scrutiny before deployment.

European values of equity further guide AI use. Policymakers stress avoiding "health data poverty" where AI benefits only data-rich populations [[Schmidt et al. 2024](#), *Mapping the regulatory landscape for artificial intelligence in health within the European Union*, 143]. There is active interest in ensuring AI algorithms do not exacerbate disparities. The EU's legally binding Charter of Fundamental Rights and healthcare equity norms mean member states monitor algorithmic bias and require inclusive design. In contrast to U.S. market-driven pilots, EU programs often start with publicly funded research consortia and pilot networks. For example, the newly launched European Health Data Space (EHDS) aims to enable cross-border data sharing under GDPR to train AI for population health, but such initiatives advance gradually due to legal and ethical review.

In summary, EU AI in healthcare is advancing under a policy umbrella that prioritizes trustworthiness and rights. Concrete examples include AI in radiology (CE-marked imaging software in Germany and Sweden),

national AI screening programs (e.g. AI triage chatbots in the Netherlands), and EU-wide research projects (Horizon Europe pilots for AI in cardiology or oncology). Yet, rollout is often phased and accompanied by guidelines. A 2024 review notes that the EU AI Act's horizontal framework "aims to mitigate the potentially harmful effects of AI on health and well-being" [[Schmidt et al. 2024](#), *Mapping the regulatory landscape for artificial intelligence in health within the European Union*, 143], reflecting Europe's collective stance.

6.3 Regulatory and Contextual Differences

GDPR covers all personal data (health data is a sensitive category) with strict consent and breach rules. HIPAA applies only to health providers and insurers and allows certain data uses without explicit consent for care. Crucially, GDPR applies globally to any entity processing EU residents' data, which even affects U.S. companies. GDPR fines (up to 4% of global turnover) are much steeper than HIPAA's capped penalties. In practice, U.S. hospitals can often use de-identified data more freely for AI, whereas EU institutions must navigate patient consent or strict anonymization.

Europe has an across-the-board AI law with classifications and standards. All "AI-enabled medical devices" in the EU are deemed high-risk under the Act, invoking stringent requirements (e.g. human oversight, transparency). Providers must implement AI Quality Management Systems and document algorithms in detail. By contrast, the U.S. has no dedicated AI law; AI tools fall under existing FDA oversight or healthcare guidelines. The FDA emphasizes patient outcomes and adaptive learning, but does not impose an EU-style risk taxonomy. U.S. policy tends to be sectoral (device/diagnostic rules, plus FDA's Nascent Good ML recommendations) rather than an omnibus AI code.

U.S. hospitals compete in a largely private market and often fund AI to gain competitive advantage. This is reflected in venture investments and hospital IT initiatives. European health systems are mostly public or single-payer, so AI projects often emerge from government or consortium funding. The European Commission's AI strategy and funding calls (e.g. for the EHDS) explicitly aim to spur "responsible AI development" in healthcare. Thus, the pace and locus of innovation differ: many U.S. trials are driven by hospital CIOs and startups, whereas EU trials are frequently aligned with public health priorities.

EU law embeds precaution: under MDR/IVDR, clinical AI tools face pre-market review and strict post-market surveillance. Proposed EU rules (AI Act) will classify medical AI as "high risk", requiring rigorous conformity assessments. GDPR grants EU patients' rights to understand AI-driven decisions. The US approach has been more "permissionless": the FDA collaborates internationally (IMDRF) on SaMD guidance [[Pesapane et al. 2021](#), *Legal and Regulatory Framework for AI Solutions in Healthcare in EU, US, China, and Russia: New Scenarios after a Pandemic*, 125] and is developing pathways for adaptive algorithms [[Leivaditis et al. 2025](#), *Artificial Intelligence in Thoracic Surgery: A Review Bridging Innovation and Clinical Practice for the Next Generation of Surgical Care*, 97], but no single US law governs AI. HIPAA addresses data security, but with limited scope. The upshot is that Europe may have stronger safety guardrails (at the cost of slower rollout), while the US leads in volume of AI devices cleared but still works to fill ethical/regulatory gaps. Both sides continue to update policies: for example, FDA's new initiatives for continuous-learning AI, and EU's AI regulation proposals. The AI Act forbids certain practices (social scoring) and imposes liability on AI providers for harms. The U.S. relies more on professional standards and malpractice law; there is no blanket ban on specific AI applications. 40% of U.S. survey respondents did note "regulatory uncertainty" as a barrier [[Poon et al. 2025](#), *Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges*, 127], reflecting this gap. In Europe, that uncertainty is replaced by clear (if burdensome) rules.

Coming into illustrative use cases to substantiate our claim regarding US and EU, for instance, ambient transcription tools (like Ambient Clinical Notes) are universally used among large health systems [[Poon et al. 2025](#), *Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges*, 127]. Another example is AI-assisted radiology: tools to flag lung nodules or brain bleeds have FDA approval and are deployed in ERs. Machine learning is also used for hospital management: AI-driven scheduling systems optimize ICU staffing, and natural-language AI automates coding and billing. During the COVID-19 pandemic,

U.S. public health agencies even used AI to project case trends (albeit with mixed success). In each case, the incentive is concrete: reduce costs or improve outcomes. A Becker's report notes that 53% of U.S. health systems rate clinical documentation AI as "high success" and 77% cite immature tools as their main hurdle.

In Europe, AI is visible in pilot programs and specialized applications. Scandinavian hospitals use AI for EEG and ECG analysis to detect neurological events. Spain and Poland have explored AI-driven digital triage for patients before hospital admission. The EU-funded EIT "AI in Health" initiative lists dozens of pilot projects, e.g., deep-learning analysis of dermatology images in Finland, or AI chatbots in the UK NHS (UK, though not an EU member, follows similar GDPR constraints). In Germany, a model project at a university clinic uses AI to recommend personalized cancer treatments based on genomic data, under strict data governance rules. Even EU-wide initiatives like the European Reference Networks encourage AI research on rare diseases. These projects tend to be carefully monitored for bias and privacy compliance, reflecting the EU's emphasis on health equity and human-centric design.

AI in healthcare is advancing on both sides of the Atlantic, but under different paradigms. The U.S. landscape prioritizes hospital-led innovation for efficiency and patient care gains (often validated in real world use)

[[Medscape Medical News](#), McKenna 2025, *Doctors Warm to AI Future in Healthcare, AMA Survey Finds*, 104].

Europe's approach is more cautious and principled: AI adoption is guided by comprehensive laws (GDPR, EU AI Act) and values that emphasize patient rights, data stewardship, and equitable outcomes. As one analysis notes, EU policy sets a "baseline regulatory framework" for health AI even before sector-specific AI laws take full effect. Ultimately, both regions see AI as transformational, but they balance innovation and protection differently. Comparing their experiences we see that U.S. tech-driven breakthroughs versus EU's risk-based rollout, offers valuable lessons for ensuring AI improves healthcare while safeguarding people and values across diverse systems.

Chapter 7: The Indian Imperative: Tailoring AI Governance for Healthcare Transformation

7.1 The Indian Context: A Nation of High Potential and Pressing Needs

India presents a unique and compelling case for the strategic deployment of artificial intelligence in healthcare. The nation's socio-economic profile is one of stark contrasts: it is one of the world's largest economies, with a projected nominal GDP of approximately \$4.19 trillion in 2025, ranking it fourth globally [[IMF Dataset](#), 79]. However, this macroeconomic strength is juxtaposed with a low per capita income of around \$2,880, which places it significantly behind other major economies and indicates vast internal disparities in wealth and access to resources. This combination of high aggregate potential and widespread individual need creates both the opportunity and the urgency for transformative solutions.

The Indian healthcare system reflects these contrasts and is beset by several long-standing challenges. A critical issue is the severe **rural-urban disparity** in healthcare infrastructure and personnel. While urban centers have a concentration of advanced medical facilities and specialists, rural areas, where the majority of the population resides, are chronically underserved. In 2022, over 80% of healthcare professionals were located in urban areas, serving only about 32% of the population [[Drishti IAS Report](#), 2025, *Transforming India's Healthcare System*, 46]. This mal-distribution is also evident in the **doctor-patient ratio**. While national averages may appear to approach the World Health Organization (WHO) recommendation of 1:1000, the reality on the ground is a significant shortage, particularly in rural regions [[Deccan Herald Report](#), Ray 2025, *Union Budget 2025 | High cost of medical education forces students to pursue studies abroad: Economic Survey*, 136]. [[Kumar and Pal 2018](#), *India achieves WHO recommended doctor population ratio - A call for paradigm shift in public health discourse!*, 91] notes India has met the WHO's 1:1000 doctor-to-population target (about 1.3 doctors per 1000 people), but rural areas suffer far fewer specialists, and many citizens lack timely care. In this context, AI-driven healthcare tools could be transformative by extending scarce expertise into remote regions. For example, India's National

Telemedicine Service (eSanjeevani) has already delivered millions of consultations, illustrating the potential of digital health to bridge gaps.

Affordability remains a major barrier to care for a large segment of the population. Out-of-pocket expenditure (OOPE) on health remains high, at around 47.1% in FY23, pushing many families into poverty due to medical costs [[Drishti IAS Report](#), 2025, *Transforming India's Healthcare System*, 46]. Government initiatives like the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY) have expanded insurance coverage to millions, but low overall insurance penetration and high costs in the largely unregulated private sector continue to pose challenges. Furthermore, the **quality of care** is inconsistent across both public and private facilities. Public hospitals often suffer from overcrowding and resource shortages, while the private sector can be prohibitively expensive and lacks uniform standards of care. These systemic issues like access, affordability, and quality make a compelling case for leveraging AI as a force multiplier to bridge gaps and improve health outcomes for all Indians.

7.2 AI Applications in Indian Healthcare: Case Studies and Legal Implications

India's healthcare system faces acute resource and access challenges. India has only about 0.76 doctors per 1,000 people (compared to WHO's 1.0 benchmark) [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66], and rural populations often lack specialists. In this context, artificial intelligence (AI) is seen as a way to amplify scarce expertise and expand reach. Over 120,000 Indians have been screened with AI-enabled X-rays since 2020, leading to hundreds of additional TB diagnoses in peripheral hospitals [[Qure.ai Report](#), 2024, *Artificial intelligence game changer in tracking cases of tuberculosis*, 131]. NITI Aayog's 2018 National Strategy for AI highlighted healthcare (cancer detection, diabetic retinopathy, telemedicine) as a key focus [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66]. Market reports forecast rapid growth in Indian health AI – one analysis projects a \$25 billion boost to GDP by 2028 [[Kalra et al. 2024](#), *Advancements in AI based healthcare techniques with FOCUS ON diagnostic techniques*, 85]. Both government and private actors have launched AI initiatives in diagnostics, surgery, administration, and public health. This report examines specific Indian examples, analyzing benefits (in access, accuracy, efficiency, outcomes) and noting implementation challenges (workforce impacts, trust). We also assess regulatory developments, especially the Digital Personal Data Protection Act, 2023 – and their implications for privacy and ethics in healthcare AI.

7.2.1 AI in Clinical Diagnostics

AI-driven diagnostic tools are proliferating across India, especially in medical imaging and screening. For instance, startups like Qure.ai deploy deep-learning on chest X-rays and CT scans to detect tuberculosis, pneumonia, stroke, and other conditions [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. Reports indicate this has boosted case detection by 30–40% and flagged many asymptomatic/incidental cases that conventional screening would miss. [[Qure.ai Report](#), 2024, *Artificial intelligence game changer in tracking cases of tuberculosis*, 131]. Qure.ai's qXR system has received regulatory approval (CDSCO Class B) for chest X-ray analysis [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66], and is used in public-sector TB screening initiatives. Niramai offers a portable, radiation-free thermal-imaging device for early breast cancer detection, expanding screening to women who lack access to mammography. In ophthalmology, companies like Artelus and Remidio use AI-based fundus imaging to screen for diabetic retinopathy in community clinics [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. India's ophthalmology centers are piloting smartphone-based DR screening. For instance, one Mumbai study (n=231) using an offline AI on mobile fundus images achieved 100% sensitivity and 88.4% specificity for referable DR [[Natarajan et al. 2019](#), *Diagnostic Accuracy of Community-Based Diabetic Retinopathy Screening With an Offline Artificial Intelligence System on a Smartphone*, 115]. Earlier work with the EyeArt AI system (a cloudbased CNN) reported 95.8% sensitivity for any DR and >99% for sight-threatening DR [[Rajalakshmi et al. 2018](#), *Automated diabetic retinopathy detection in smartphone-based fundus photography using artificial intelligence*, 113]. These high accuracy figures suggest AI can overcome resource gaps (few retinal specialists) and greatly expand screening in rural areas. Such tools have improved

access to specialist-level diagnostics in underserved areas. For example, AI-based diabetic eye screening deployed in telemedicine camps has found retinopathy with accuracy comparable to ophthalmologists, enabling timely referrals.

In neurology and oncology, collaborative R&D projects have emerged. The Indian Council of Medical Research (ICMR) notes initiatives such as an AI tool at Aster CMI Hospital for carpal tunnel syndrome screening, and partnerships (IIT, Intel, Qure.ai, Medtronic) to develop algorithms for brain tumor grading and stroke detection [Kundu and Bardhan 2025, *Artificial intelligence in neurology, ethics, recent guideline, and law-an Indian perspective*, 92]. These promise faster triage and specialist support in acute care. In cardiovascular care, Wipro GE Healthcare launched an “AI-enabled cath lab” for real-time imaging analytics, and Apollo Hospitals (with Microsoft) developed predictive AI models for cardiac risk [Chettri et al. 2025, *Bridging the Gap in the Adoption of Trustworthy AI in Indian Healthcare: Challenges and Opportunities*, 28], aiding clinicians in early intervention.

AI in diagnostics has delivered notable benefits: increased efficiency, earlier case detection, and extended coverage. By automating image interpretation, it can screen large volumes of patients quickly, which is critical in high-burden settings (e.g. India’s TB epidemic). For instance, Qure.ai’s chest X-ray algorithm enables non-specialist clinics to flag TB suspects, reducing diagnostic delays [Kapoor et al. 2024, *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. Niramai’s device allows community health workers to detect occult breast lumps, improving early treatment rates in rural areas. On accuracy, studies indicate these AI tools often match or exceed human performance in pattern recognition (radiomics techniques reduce false positives in imaging [Kalra et al. 2024, *Advancements in AI based healthcare techniques with FOCUS ON diagnostic techniques*, 85]).

Impact: Faster throughput of scans, better screening outreach in remote areas, and early detection of diseases that were previously underdiagnosed.

However, real-world success is mixed. Many AI services remain limited to pilot programs or urban hospitals. For example, the da Vinci robotic surgery system (used in oncology and urology) exists in a few dozen centers [Deo and Anjankar 2023, *Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India*, 40], largely private hospitals; its high cost has prevented nationwide adoption. In screening, widespread deployment requires further validation and infrastructure (e.g., reliable internet, digital archives), which some regions lack. Overall, diagnostics AI in India has improved access and accuracy where applied, but its benefits have not yet fully reached the rural or low-resource public sector.

7.2.2 AI in Surgical Care and Robotics

Robotics and AI are increasingly integrated into surgery and hospital logistics. India saw its first da Vinci robotic surgery at AIIMS (Delhi) in 2006, and by 2019 over 66 centers performed >12,800 procedures using surgical robots [Deo and Anjankar 2023, *Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India*, 40]. These systems (used in urology, gynecology, etc.) improve precision, reduce blood loss, and shorten hospital stays. For example, robot-assisted prostatectomy and hysterectomy in tertiary hospitals have yielded outcomes comparable to open surgery but with faster recovery.

Beyond the da Vinci, AI is used in robotic-assisted interventions. In neurosurgery, AI-enabled image guidance helps plan complex tumor resections. Partnerships (e.g., Tata Memorial Hospital with tech institutes) are developing AI algorithms to identify tumor characteristics on MRI. In acute stroke care, Medtronic India and Qure.ai are collaborating on an AI tool to quickly detect stroke on scans [Kundu and Bardhan 2025, *Artificial intelligence in neurology, ethics, recent guideline, and law-an Indian perspective*, 92], potentially aiding neurosurgeons in timely clot removal.

Robotic systems also support non-surgical tasks. During COVID-19, Indian hospitals deployed “hospitality” robots (e.g. SoftBank’s Pepper) for tasks like patient screening, delivering medications and meals, and disinfecting rooms. These reduce health-worker exposure and free staff for critical duties. Automated guided

vehicles (AMRs) have performed functions of nurses or orderlies like carrying supplies, guiding patients – especially where infection control is paramount. Such AI-robotic integration bridges workforce gaps, performing routine jobs in places with few nurses, and easing workload [[Deo and Anjankar 2023](#), *Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India*, 40].

Examples: Da Vinci surgical robots (private and govt hospitals); neurosurgical AI models (IIT Roorkee, AIIMS); COVID-19 screening and assistant robots (Kerala, other states).

Impact: Enhanced surgical precision and efficiency; reduced procedure times; improved infection control via automated screening/disinfection robots; support for staff shortages.

Drawbacks/Challenges: Robotic platforms are expensive and require extensive training. Many Indian hospitals view them as high-end luxury (rather than essential) – one review warns of “irrational demand” where hospitals acquire robots for prestige rather than patient need [[Deo and Anjankar 2023](#), *Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India*, 40]. There are also medico-legal concerns: surgical robots could fail or be hacked, yet liability is unclear. Physiotherapy and elderly-care robots are nascent; cultural acceptance of robots in personal care varies. The introduction of robots may reduce the “human touch” in care, an issue of trust and empathy. Overall, AI/robotics in surgery has benefited precision and access to specialty care in centers that can afford it, but cost and training remain barriers.

7.2.3 AI in Hospital Administration and Management

AI is streamlining the “behind-the-scenes” aspects of hospital operations. Administrative workload like data entry, record-keeping, and resource management – is increasingly assisted by AI and software automation. For example, Eka Care (a private EMR startup) uses OCR and natural-language processing to auto-digitize doctors’ handwritten notes and link them to the government-issued Ayushman Bharat Health Account (ABHA) IDs [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. It also provides patients a secured “Digilocker” for medical histories. This enables teleconsultation with complete histories, aiding continuity of care. Such applications promise greater efficiency in India’s often fragmented healthcare data ecosystem by reducing manual paperwork, improving data interoperability, and giving clinicians instant access to past records.

“Robotic process automation” (RPA) software is another administrative AI trend: simple “software robots” can mimic human clerks to pull data, process claims, or manage appointments [[Deo and Anjankar 2023](#), *Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India*, 40]. Many health insurers and hospitals are piloting AI chatbots for patient engagement for appointment scheduling or FAQs, though large-scale examples in India are still emerging.

At the policy level, India’s Ayushman Bharat Digital Mission (ABDM) promotes standardized digital health records and telemedicine. Its interoperability frameworks (e.g. FHIR, SNOMED-CT, ICD codes) are being adopted by hospitals. Under ABDM’s Health Data Management Policy, institutions must employ encryption, access controls, and report breaches for all health data [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66]. As a result, even AI-based tools in approved facilities will operate within these security protocols. For instance, some state governments and the Ayushman Bharat insurance scheme (PM-JAY) now use AI to detect fraud in claims and to verify patient eligibility [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. This improves efficiency and fairness in large public programs.

Examples: Eka Care EMR (automation of patient records); RPA for insurance claims (various providers); AB-PMJAY’s AI for fraud detection; eSanjeevani telemedicine platform (MoHFW) enabling remote consults across rural clinics.

Impact: Reduced paperwork and duplication; faster patient registration and billing; earlier fraud detection; and significant expansion of teleconsultation (eSanjeevani logged millions of virtual visits during COVID, bridging rural–urban doctor gaps [[Chettri et al. 2025](#), *Bridging the Gap in the Adoption of Trustworthy AI in Indian Healthcare: Challenges and Opportunities*, 28]).

However, change is not uniform. Many smaller clinics lack even basic EMRs. Implementing AI requires training IT staff and ensuring data quality, which some facilities find daunting. There is also clinician apprehension: doctors worry AI-driven scheduling or diagnosis reminders may impinge on their autonomy or audit every decision. Trust in automated systems (especially for billing or triage) grows slowly. Nonetheless, where adopted, AI has streamlined hospital workflows, cut costs (fewer manual clerks needed), and expanded services (telehealth) that were previously impractical.

7.2.4 AI in Public Health and Population Initiatives

AI’s role in public health in India spans disease surveillance, epidemic control, and large-scale screenings. During COVID-19, AI tools supported containment efforts: the government’s eSanjeevani telemedicine (supported by rural Common Service Centres) used AI/IoT for remote monitoring and consultations. This platform proved invaluable in reaching rural patients when in-person care was disrupted, effectively “bridging the rural–urban health gap” [[Chettri et al. 2025](#), *Bridging the Gap in the Adoption of Trustworthy AI in Indian Healthcare: Challenges and Opportunities*, 28]. The government also leveraged data analytics (a basic form of AI) to optimize vaccine distribution and track hotspots.

Beyond COVID, targeted AI projects are underway for endemic diseases. NITI Aayog funded AI-based chest X-ray screening for tuberculosis in high-burden areas. In one public–private example, Wadhvani AI deployed a machine-learning model to predict “loss to follow-up” and mortality risk in TB patients, allowing health workers to intervene with those likely to drop treatment [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. This improves treatment success rates by focusing limited outreach on at-risk patients.

AI is also used in environmental health and epidemiology. While India-specific systems are nascent, research draws on global AI methods: e.g., algorithms scanning internet and social media for outbreak signals (akin to HealthMap and Google Flu Trends). The MoHFW and NITI Aayog are even building a 20,000+ image repository of cancer pathology and radiology (for major cancers) [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86], to train AI models for cancer screening – an ambitious Precision Public Health initiative.

Examples: eSanjeevani (AI-enhanced telehealth); AI-driven TB initiatives (NITI Aayog X-ray screening, Wadhvani predictive model); national health data banks for AI research.

Impact: Enhanced surveillance (early outbreak detection), personalized follow-up in TB programs, and vastly increased teleconsultation coverage. These tools help prioritize limited public resources and improve outcomes (e.g. AI-flagged TB cases can be treated earlier, reducing transmission). These examples, though still in research or pilot stages, point to AI’s growing role in Indian public health planning.

Challenges in the public-health realm include data gaps: AI models require comprehensive, high-quality data, but rural health records are often incomplete. Privacy concerns are heightened when using AI on population data. For instance, ethical debates arose when anonymized datasets (e.g. COVID app logs) risked re-identification [[Naithani 2023](#), *COMMENTARY: Protecting healthcare privacy: Analysis of data protection developments in India*, 114]. Ensuring community trust is vital: any AI-driven policy (even beneficial) must be transparent to avoid fears of surveillance.

Overall, studies and reports emphasize that Indian applications are showing promise. A recent expert review notes “encouraging results” in AI-enabled identification of diseases like TB, Parkinson’s, pneumonia and

cancers [[Kapoor et al. 2024](#), *AI Horizons in Indian Healthcare: A Vision for Transformation and Equity*, 86]. Governmental bodies (e.g. NITI Aayog) project that AI could significantly improve healthcare reach and efficiency if well-implemented. However, most of the cited successes are still at pilot or demonstration level; large-scale, published outcome studies in the Indian context remain few.

7.2.5 Legal and Regulatory Framework

India's legal infrastructure for health data and AI is under development. Until 2023, patient privacy was mainly protected under the Information Technology Act, 2000 and 2011 "SPDI Rules" [[Tandon and Gupta 2025](#), *Informational Privacy in the Age of Artificial Intelligence: A Critical Analysis of India's DPDP Act, 2023*, 148]. Those rules defined certain health attributes as "sensitive" and required consent for data collection, but enforcement was weak (they applied only to "body corporates") [[Naithani 2023](#), *COMMENTARY: Protecting healthcare privacy: Analysis of data protection developments in India*, 114]. Notably, breaches did occur – for example, a Maharashtra lab leaked over 35,000 patient records in 2016 [[Bajpai and Wadhwa 2021](#), *Artificial Intelligence and Healthcare in India*, 14] – illustrating risks in the absence of strong safeguards.

The Digital Personal Data Protection Act, 2023 (DPDP Act) finally provides a comprehensive data law. It mandates that healthcare providers obtain explicit, informed consent for personal health data [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66], and it creates a new "Consent Manager" role to let individuals review and withdraw consent. The Act (effective 2023) will eventually impose obligations on data fiduciaries and penalties for breaches. However, analysts note important gaps: DPDP does not define "sensitive personal data" or grant rights like data portability, and it allows some data processing without explicit consent. Sector-specific frameworks (e.g. the draft Digital Information Security in Healthcare Act (DISHA) and the 2022 Health Data Management Policy) envision stricter rules – such as requiring patient-centric controls and audit trails which DPDP lacks [[Naithani 2023](#), *COMMENTARY: Protecting healthcare privacy: Analysis of data protection developments in India*, 114]. Thus, there is uncertainty about which standard applies if the health policy and DPDP diverge.

For AI tools specifically, the Central Drugs Standard Control Organization (CDSCO) now regulates certain software as medical devices (SaMD). For instance, Qure.ai's qXR received official approval under medical device rules, meaning any significant AI software change requires regulatory clearance. The Ayushman Bharat Digital Mission (ABDM) also enforces standards: under its Health Data Management policy, digital health data (including AI-generated data) must be encrypted and breach-reported, and AI algorithms are expected to be fair and explainable. International research in India must pass BioRRAP/SCRIP review, ensuring foreign AI studies comply with ethical norms [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66].

Existing rules: IT Act 2000/SPDI Rules (limited), ICMR ethical guidelines (non-binding) [[Naithani 2023](#), *COMMENTARY: Protecting healthcare privacy: Analysis of data protection developments in India*, 114], and professional standards (telemedicine guidelines).

DPDP Act 2023: First broad data law; requires consent and imposes fiduciary duties, but critics call for explicit human oversight of automated decisions [[Tandon and Gupta 2025](#), *Informational Privacy in the Age of Artificial Intelligence: A Critical Analysis of India's DPDP Act, 2023*, 148].

Ayushman Bharat/NDHM: Interoperability standards (FHIR, ICD-10) and data protection norms (HDM policy) for all digital health platforms.

Upcoming: Proposed DISHA Act (still pending) would set even stricter health data rules, and a new Digital India Act (in draft) would overhaul cyber law to include AI liability [[Gupta et al. 2025](#), *Legalities, Ethics and Frameworks for Healthcare AI in India*, 66].

7.2.6 Effects and Fallouts

The previous regulatory vacuum has had tangible consequences. Analyses of Indian healthcare data practices have documented frequent privacy lapses – e.g. clinics sharing patient data with researchers without anonymization, or failing to notify patients of breaches [[Naithani 2023](#), *COMMENTARY: Protecting healthcare privacy: Analysis of data protection developments in India*, 114]. AI amplifies these risks: a report notes that de-identified health records analyzed by AI can often be re-identified, potentially exposing sensitive health status. Until clear rules and enforcement exist, patients and providers operate in a gray zone. For instance, without a mandated “right to be forgotten,” data used by an AI in a research context might be retained indefinitely. The DPDP Act and NDHM policies start to address this, but experts urge more: one critic argues DPDP should require human review of critical automated decisions [[Tandon and Gupta 2025](#), *Informational Privacy in the Age of Artificial Intelligence: A Critical Analysis of India’s DPDP Act, 2023*, 148].

Despite the potential, India’s legal framework for AI in healthcare is underdeveloped, creating risks of “fallout” if issues go unchecked. Notably, experts observe that India “does not explicitly address telehealth” in existing laws, nor are there robust data protection rules for health information [[Jain 2023](#), *Regulation of Digital Healthcare in India: Ethical and Legal Challenges*, 83]. In practical terms, this means patient privacy can be compromised and there is no clear process for obtaining informed consent when using digital health tools. In September 2023 India enacted the Digital Personal Data Protection Act (DPDP Act), extending privacy protections to “digital personal data” [[International Bar Association Report](#), Jindel and Gupta, 2023, *AI in healthcare: trends and challenges in India*, 78], but its adequacy for sensitive medical data is yet to be tested.

Equally concerning is liability for AI errors. If an AI diagnostic tool gives a wrong result, who is responsible – the physician, the software developer, or the hospital? As one legal analysis notes, questions about accountability for AI-driven decisions “remain largely unanswered due to lack of laws” in India. For example, it is not yet clear whether a doctor can legally override an AI diagnosis or must follow it, or how to compensate a patient harmed by an algorithm. These gaps could deter adoption and leave patients without recourse.

Additionally, algorithmic bias and patient safety are unresolved concerns. India’s diverse population means AI trained on non-representative data might misdiagnose under-served groups. High-profile AI failures elsewhere – such as a medical chatbot suggesting suicide [[International Bar Association Report](#), Jindel and Gupta, 2023, *AI in healthcare: trends and challenges in India*, 78] – highlight that autonomous algorithms may give “erratic and unpredictable” advice if not carefully constrained. Without strong validation protocols and transparency (e.g. explaining AI reasoning), the risk of harm persists.

Experts therefore urge creation of ethical/regulatory frameworks for medical AI. Indian scholars recommend updating medical device laws and malpractice rules to explicitly cover software and AI tools. Professional guidelines for physicians on AI use – including validation, consent, and data security – are also being called for. In summary, while India is embracing AI innovations in healthcare, the absence of a comprehensive legal and ethical framework remains a significant challenge [[Jain 2023](#), *Regulation of Digital Healthcare in India: Ethical and Legal Challenges*, 83] [[International Bar Association Report](#), Jindel and Gupta, 2023, *AI in healthcare: trends and challenges in India*, 78].

Worldwide, AI has demonstrated revolutionary improvements in disease detection and medical workflow (from skin cancer classification to ECG analysis) [[Esteva et al. 2017](#), *Dermatologist-level classification of skin cancer with deep neural networks*, 50] [[Rajpurkar et al. 2017](#), *CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning*, 134]. In India, early pilots in TB and eye-care screening show tangible benefits (e.g. higher case detection). However, translating these technical successes into routine care requires addressing data privacy, liability and bias. Current Indian law provides few clear answers: telemedicine and AI-enabled care are essentially operating in a legal grey zone. The coming years will test whether India can adapt its regulations to safeguard patients while harnessing AI’s potential, so that these powerful tools improve healthcare delivery without unintended “fallouts.” In short, India’s health AI ecosystem is emerging faster than its laws, raising ongoing ethical and legal challenges.

7.3 India's Data Protection Cornerstone: The Digital Personal Data Protection (DPDP) Act, 2023

In August 2023, India enacted its first comprehensive, cross-sectoral data privacy law, the Digital Personal Data Protection (DPDP) Act [[Burman 2023](#), *Understanding India's New Data Protection Law*, 20]. This law enshrines GDPR-like principles: data must be processed lawfully, fairly and transparently; collected only for specified purposes; limited to what is necessary; kept accurate and for a limited time; and protected by strong security safeguards. In healthcare, this means patient information used by AI (e.g. X-rays, genetic data) must respect consent, purpose and security requirements as under DPDP. Non-digital data (e.g. paper records) are currently outside DPDP's scope, highlighting areas for future regulation. India also lacks a dedicated AI law: government policy relies on broad strategy documents (National AI Strategy 2018) and sectoral guidance. As one legal analysis confirms, "currently, there are no specific codified laws... in India that directly regulate AI" [[White & Case Report](#), 2024, *AI Watch: Global regulatory tracker – India*, 160]. This landmark legislation establishes the foundational legal framework for data governance and must be considered the non-negotiable cornerstone upon which any future AI regulation is built. The DPDP Act applies to the processing of digital personal data within India and has extraterritorial reach, covering the processing of data of Indian residents by foreign entities for the purpose of offering goods or services in India [[PRS India Report, Ministry of Electronics and Information Technology](#), 130].

India's Digital Personal Data Protection Act (DPDP Act, 2023) is newer and narrower. It covers only digital personal data; non-digital records or paper files remain outside its scope [[DPDP Act 2023, Section 2\(n\), Section 3\(1\), Meity](#), 45]. Unlike the GDPR, the DPDP Act does not distinguish "sensitive" categories, so health or educational data get no extra protection in theory; all digital personal data must be processed lawfully. Consent is generally required except in limited cases (employment, emergencies). Importantly, the DPDP Act omits a right analogous to GDPR's Article 22: there is no explicit right for a data principal to refuse automated decision-making [[DPDP Act 2023, Sections - 6\(4\), 11\(1\), 12\(1\), 12\(3\), 14\(1\), Meity](#), 45]. DPDP does give data principals rights to access, correction, erasure and to revoke consent, and even a unique right to appoint someone to exercise their rights after death. It also mandates that institutions handling children's data obtain parental consent and forbid processing harmful to children [[SS Rana & Co. Article](#), 2025, *Ensuring Student Privacy in Education: Compliance with the DPDP Act, 2023*, 147]. In practice, many Indian schools and hospitals must navigate both the DPDP Act (once in force) and existing IT Act rules or sectoral regulations (like the National Medical Commission's codes).

The Act is fundamentally **consent-centric**. It mandates that personal data can only be processed for a lawful purpose after obtaining the "free, specific, informed, unconditional and unambiguous" consent of the individual, signified by a clear affirmative action. A clear and plain-language notice detailing the data to be collected and the purpose of processing must precede the request for consent [[CookieYes Article](#), Safna 2025, *India's DPDP Act Explained: The Latest Guide for Compliance*, 35]. However, the Act also introduces the concept of "**legitimate uses**" for which consent is not required. These include instances where an individual voluntarily provides data, for the provision of state benefits and services, medical emergencies, public health threats, and employment-related purposes [[PRS India Report, Ministry of Electronics and Information Technology](#), 130].

The DPDP Act grants several rights to individuals, referred to as **Data Principals**. These include the right to access a summary of their data, the right to correct or erase their data, and the right to grievance redressal [[Burman 2023](#), *Understanding India's New Data Protection Law*, 20]. Entities that determine the purpose and mean of processing data, known as **Data Fiduciaries**, have corresponding obligations. They must ensure data accuracy, implement reasonable security safeguards to prevent data breaches, delete data once its purpose is fulfilled, and notify the Data Protection Board of India and affected individuals in the event of a breach [[PRS India Report, Ministry of Electronics and Information Technology](#), 130]. The Act also introduces heightened obligations for **Significant Data Fiduciaries (SDFs)**, which are designated based on the volume and sensitivity of data they process, requiring them to appoint a DPO and conduct data protection impact assessments [[DPDP Act 2023, Sections - 10\(1\), 10\(2\)\(a\), 10\(2\)\(c\)\(i\), 10\(2\)\(b\), 10\(2\)\(c\)\(ii\), Meity](#), 45].

Crucially for the healthcare sector, the Act includes stringent provisions for processing the data of children (defined as individuals under 18 years of age), requiring verifiable parental consent and prohibiting tracking, behavioral monitoring, or processing that is likely to cause any detrimental effect [[DPDP Act 2023, Sections - 9\(1\), 9\(2\), 9\(3\), Meity, 45](#)]. The establishment of the Data Protection Board of India to adjudicate non-compliance and impose significant financial penalties (up to ₹250 crore) provides the law with an enforcement mechanism [[Burman 2023, Understanding India's New Data Protection Law, 20](#)]. The DPDP Act thus creates a definitive legal standard for privacy that all AI applications processing personal data in India must respect.

Although not yet fully in force, DPDP is shaping expectations. Indian universities and hospitals dealing with digital records will need to appoint data protection officers and seek consent for processing personal data. In healthcare, medical records maintained in digital form will fall under DPDP: hospitals must ensure data principals can access and correct their records, and can withdraw consent. However, since DPDP lacks a prohibition on automated decisions, an AI-driven diagnostic tool could in theory be used without offering patients the specific right to object (a gap compared to GDPR).

To enable AI in healthcare without undermining privacy, India should pursue an AI framework aligned with DPDP. In practice, this could mean adapting elements of the EU model: for instance, classifying healthcare AI as high-risk (subject to oversight) while explicitly deferring to DPDP for personal data issues. Such an approach would protect India's data sovereignty (ensuring health AI respects Indian patients' rights) yet allow innovation. It could also leverage India's strong IT sector: by setting clear standards (risk management, impact assessments) Indian AI developers could become globally competitive, while giving assurances to citizens that their health data are safeguarded.

7.4 The Case for an Indian AI Act in Healthcare: Opportunities and Challenges

The unique confluence of India's pressing healthcare needs and its rapidly expanding digital infrastructure presents an unparalleled opportunity for AI-driven transformation [[Kapoor et al. 2024, AI Horizons in Indian Healthcare: A Vision for Transformation and Equity, 86](#)]. AI can directly address the system's core challenges. For instance, AI-powered telemedicine and diagnostic tools can help bridge the rural-urban access gap, bringing specialist-level expertise to remote areas [[Makani et. al 2024, Artificial Intelligence-powered Healthcare for India: Promises, opportunities and challenges, 99](#)]. AI can optimize resource allocation in overburdened public hospitals, improve the accuracy of disease surveillance, and accelerate the development of affordable drugs for diseases prevalent in India [[World Economic Forum Article, Ghewade 2025, 4 ways India is deploying AI and innovation to revolutionize healthcare, 61](#)].

However, the path to AI adoption is fraught with challenges specific to the Indian context. These include a shortage of specialized AI talent, the complexity of collecting and digitizing high-quality data from a diverse population, and the immense difficulty of scaling solutions across varied socio-economic and infrastructural landscapes [[Makani et. al 2024, Artificial Intelligence-powered Healthcare for India: Promises, opportunities and challenges, 99](#)].

India already has promising AI healthcare projects. Notably, during the COVID-19 pandemic, IISc Bangalore (with DST support) developed XraySetu, an AI platform for early COVID screening. XraySetu allows rural doctors to send low-resolution chest X-rays via WhatsApp and receive an AI-generated report highlighting lung abnormalities [[Ministry of Science and Technology Report, GoI, 2021, A new AI-driven platform will facilitate early-COVID interventions over Whatsapp, 64](#)]. Tested on over 125,000 UK images and 1,000+ Indian patients, it achieved ~98.9% sensitivity. The system uses transfer learning to work even with nonCOVID lung images, and provides annotated heatmaps for physician review. According to its developers, XraySetu – a collaboration of academia and industry has already served over 1,200 cases in rural India. Its creators highlight that such technology “can enable cutting-edge AI-driven systems powering mobile PHCs, which can make healthcare more accessible... at a fraction of the cost” [[Ministry of Science and Technology Report, GoI, 2021, A new AI-driven platform will facilitate early-COVID interventions over Whatsapp, 64](#)]. This example shows how AI tools,

when integrated into local workflows (WhatsApp chatbots, mobile clinics), can help overcome doctor shortages and limited infrastructure.

Another IISc effort, AnamNet, demonstrated AI on chest CT scans. Developed in 2021, AnamNet uses deep learning to segment lung images and estimate COVID-19 damage. It provides rapid automated analysis to assist doctors with diagnosis and patient management. Notably, the team also built a mobile app (CovSeg) that runs the model on a smartphone, enabling point-of-care use without expensive hardware. These innovations like XraySetu for X-rays and AnamNet/CovSeg for CT scans illustrate how Indian researchers are tailoring AI to local needs (low resources, mobile use) while maintaining accuracy [[Kernel, IISc Report](#), Bhattacharya 2021, *AI-based Tool for Automated Diagnosis of COVID-19 Lung Infection*, 18].

In a typical hospital, each patient has an electronic health record containing demographics, clinical history, medications, lab results, and images. Wearable devices and mobile apps add continuous monitoring data (heart rate, glucose, fitness). Public health databases and research cohorts contribute large-scale, de-identified datasets. To train AI, these heterogeneous data are often uploaded to cloud servers or shared research repositories [[Yadav et al 2023](#), *Data Privacy in Healthcare: In the Era of Artificial Intelligence*, 165]. For example, a federated learning study might aggregate thousands of chest X-rays from different hospitals (often de-identified first) into one dataset. The above-mentioned homecare OCT system specifically sends patient eye images to a cloud-AI for analysis [[Bende et. al 2022](#), *A Case Study on Data Protection for a Cloud- and AI-based Homecare Medical Device*, 17]. In each case, the collection methodology must adhere to privacy laws: for instance, Indian rules will require that a hospital inform patients how their data will be used in AI training and may need explicit consent if beyond direct care.

Across both domains, GDPR and DPDP impose data security and storage requirements. Educational and health data must be adequate, relevant, and limited to the purposes. GDPR's data minimization principle conflicts with AI's "data-hungry" nature, pushing designers to use only necessary data or anonymize it. Institutional review boards and ethics committees play a role in approving data sharing for research; for example, retrospective AI studies in medicine sometimes get IRB waivers if data are fully anonymized [[Yadav et al 2023](#), *Data Privacy in Healthcare: In the Era of Artificial Intelligence*, 165]. Open-source datasets (like TCIA) illustrate one compliance strategy: the data providers have taken patient identifiers out, though, as experts warn, future AI may re-link them.

While the DPDP Act provides a vital framework for data privacy, it is necessary but insufficient to govern the multifaceted risks of AI in healthcare. The DPDP Act's mandate is to protect personal data; it is not designed to regulate the safety, clinical efficacy, or ethical integrity of an AI algorithm itself [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113]. An AI diagnostic tool could be fully compliant with the DPDP Act using properly consented data and employing robust security measures, yet still be medically unsafe due to clinical inaccuracies or systematically biased against certain Indian demographics. This creates a significant regulatory lacuna. The risks posed by healthcare AI extend far beyond privacy to encompass patient safety, clinical validity, and equity, which require a specialized regulatory lens [[Chettri et al. 2025](#), *Bridging the Gap in the Adoption of Trustworthy AI in Indian Healthcare: Challenges and Opportunities*, 28].

Therefore, a dedicated, sector-specific AI in Healthcare Act is imperative for India. Drawing lessons from the global paradigms, such an Act would not replace the DPDP Act but would work in concert with it. It would address the non-privacy risks unique to healthcare AI, establishing clear standards for validation, performance monitoring, bias audits, and accountability. This dual-regulatory approach is essential to build trust among patients and clinicians, encourage responsible innovation, and ensure that AI is deployed in a manner that genuinely serves India's public health goals. Furthermore, India's advanced Digital Public Infrastructure (DPI), including Aadhaar and the Ayushman Bharat Digital Mission, creates a unique environment [[World Economic Forum Article](#), Ghewade 2025, *4 ways India is deploying AI and innovation to revolutionize healthcare*, 61]. While this interconnected infrastructure offers an unprecedented opportunity to train large-scale AI models on national-

level data, it also dramatically amplifies the systemic risks of privacy breaches and the rapid, widespread propagation of algorithmic harm. This high-stakes context makes a robust, proactive, and specialized AI regulatory framework not just advisable, but absolutely essential.

Chapter 8: A Proposed Framework for India's AI in Healthcare Act

8.1 Core Principles: Upholding the Sovereignty of the DPDP Act

The foundational principle of a proposed AI in Healthcare Act for India must be its harmonious coexistence with and subordination to the Digital Personal Data Protection (DPDP) Act, 2023. The DPDP Act must remain the sovereign law governing all aspects of personal data processing, including the data used to train and operate AI systems. The new AI Act would not create alternative data processing standards but would instead focus on regulating the AI system itself its design, validation, deployment, and impact. This creates a clear demarcation of regulatory authority: the DPDP Act governs the data, while the AI Act governs the algorithm.

This proposed framework can be strengthened by drawing upon a comparative analysis of the DPDP Act and the GDPR. While both are rooted in similar privacy principles, the DPDP Act has unique features tailored to the Indian context that must be central to the AI Act's design [[Legal500 Report](#), 2024, *GDPR V INDIA'S DPDPA: KEY DIFFERENCES AND COMPLIANCE IMPLICATIONS*, 96]. For instance, the DPDP Act's consent-centric model, which requires consent to be "free, specific, informed, unconditional and unambiguous," is stricter than the GDPR's allowance for other lawful bases like "legitimate interests". The AI Act must respect this high bar for consent when AI systems process personal data. Similarly, the DPDP Act's higher age of consent for children (18 years) and its explicit prohibition on tracking and behavioral monitoring of minors must be strictly enforced for any pediatric healthcare AI applications [[Latham & Watkins 2023](#), *India's Digital Personal Data Protection Act 2023 vs. the GDPR: A Comparison*, 95]. The AI Act should also integrate the DPDP Act's concept of "consent managers," providing a mechanism for individuals to manage their consent for their health data being used by various AI tools through a trusted intermediary. By building upon these uniquely Indian provisions, the AI Act can ensure that it is not merely a transplant of foreign regulation but a framework that is deeply integrated with India's domestic legal philosophy.

8.2 Balancing Innovation and Regulation: A Tailored, Risk-Based Approach for India

To navigate the delicate balance between fostering a vibrant AI innovation ecosystem and protecting citizens from potential harm, the proposed AI in Healthcare Act should adopt a pragmatic, risk-based approach, learning from global precedents while adapting them to India's specific needs.

A central feature of the Act would be a **tiered risk-based classification system**, inspired by the EU AI Act but tailored to the Indian healthcare context:

- **Prohibited Risk:** This category would ban AI applications that pose an unacceptable threat to fundamental rights and patient safety. Examples could include AI for social scoring of patients, discriminatory insurance pricing based on predictive health analytics, or fully autonomous surgical systems operating without any human oversight.
- **High Risk:** This would be the most heavily regulated tier, encompassing AI systems with the potential for significant impact on a patient's health and life. This includes AI used for clinical diagnosis (e.g., analyzing medical images or pathology reports), treatment planning, risk prediction for critical illnesses, and robotic surgery assistance. Systems in this category would be subject to stringent pre-market conformity assessments to validate their clinical accuracy and safety, mandatory "human-in-the-loop" oversight, continuous post-market monitoring for performance degradation or emergent bias, and rigorous technical documentation and data governance requirements.
- **Limited Risk:** This tier would cover AI systems that interact with patients but have a lower potential for direct physical or psychological harm. Examples include AI-powered chatbots for patient scheduling, medication reminders, or providing general health information. The primary obligation for these systems

would be transparency; they must clearly disclose to users that they are interacting with an AI system, so individuals are not misled.

- **Minimal Risk:** This category would include AI applications used for internal administrative or operational purposes with no direct patient interaction or impact on clinical decisions, such as AI for optimizing hospital bed allocation or managing supply chains. These systems would be largely exempt from the AI Act's specific obligations, allowing for frictionless innovation in back-end processes.

To further encourage innovation, particularly among India's burgeoning startup ecosystem, the Act should formally establish **regulatory sandboxes** [[National Economic Forum](#), 2025, *Global Lessons for India's AI Regulation*, 117]. These would be controlled environments, supervised by a new regulatory body, where developers could test novel AI healthcare solutions on real-world data without facing the full force of regulatory compliance immediately. This would allow for iterative development and validation, enabling regulators to gain insights into emerging technologies while helping innovators bring safe and effective products to market more quickly.

Finally, the Act must codify principles of **accountability and explainability**. It should establish clear lines of legal liability in cases of AI-induced harm, apportioning responsibility between developers, healthcare providers (deployers), and clinicians (users). For all high-risk systems, the Act should mandate the use of "Explainable AI" (XAI) techniques to the greatest extent technically feasible, ensuring that the basis for an AI's recommendation can be understood, reviewed, and, if necessary, challenged by a human expert.

8.3 Voices from the Field: Indian AI Healthcare Narratives

To illustrate the practical functioning of this proposed framework, it can be applied to two distinct Indian use cases.

ISc Bangalore's XraySetu

XraySetu is an AI-driven platform developed during the COVID-19 pandemic to enable rapid screening of the disease using chest X-ray images sent via WhatsApp, designed specifically for use in rural and resource-limited settings [[Ministry of Science and Technology Report, GoI, 2021](#), *A new AI-driven platform will facilitate early-COVID interventions over Whatsapp*, 64].

- **Classification under the Proposed Act:** As a tool used for clinical diagnosis to detect COVID-19 and 14 other lung ailments, XraySetu would be unequivocally classified as a **High-Risk AI System**.
- **Regulatory Obligations:**
 1. **Pre-Market Assessment:** Before widespread deployment, the developers would need to submit the system for a conformity assessment by the designated regulatory body. This would involve validating its claimed performance metrics (Sensitivity: 98.86%, Specificity: 74.74%) on a large, diverse dataset representative of the Indian population, not just data from foreign sources like the UK's NIH [[Ministry of Science and Technology Report, GoI, 2021](#), *A new AI-driven platform will facilitate early-COVID interventions over Whatsapp*, 64]. The assessment would scrutinize the "Transfer Learning framework" to ensure it does not introduce biases.
 2. **Data Governance and Privacy:** The processing of X-ray images, which are personal health data, via a third-party platform like WhatsApp would be subject to intense scrutiny under the DPDP Act. The AI Act would require the developers to demonstrate robust end-to-end encryption and compliance with all DPDP provisions, including providing a clear notice and obtaining valid consent from the patient before the image is uploaded.
 3. **Human-in-the-Loop:** The Act would mandate that the platform's design reinforces its role as a decision-support tool, not a replacement for a clinician. The automated report sent back to the doctor must be clearly labeled as a preliminary analysis, with a final diagnosis remaining the sole responsibility of the qualified medical professional.

4. **Post-Market Monitoring:** Once deployed, the developers would be required to continuously monitor the system's performance in real-world conditions and report any degradation in accuracy or evidence of emergent bias to the regulatory authority.

AI to Improve the Doctor-Patient Ratio

This hypothetical use case considers the development of AI tools designed to alleviate the workload of doctors and bridge the specialist gap in India, particularly in rural areas. Such a system might include a triage chatbot and a preliminary diagnostic tool [[Deccan Herald Report](#), Ray 2025, *Union Budget 2025 | High cost of medical education forces students to pursue studies abroad: Economic Survey, 136*].

- **Classification under the Proposed Act:** The system would have components falling into different risk tiers, demonstrating the "regulate-the-use, not-the-tech" approach.
- The AI-powered **triage chatbot** that asks patients about their symptoms to direct them to the appropriate level of care (e.g., "consult a GP," "visit the emergency room") would be classified as **Limited Risk**. Its primary obligation would be transparency—clearly stating it is an AI and not a doctor.
- The **preliminary diagnostic tool** that analyzes the reported symptoms and suggests a potential diagnosis (e.g., "symptoms are consistent with viral fever") would be classified as **High Risk**.
- **Regulatory Obligations:**
 1. **Validation and Bias Mitigation:** The high-risk diagnostic component would require rigorous clinical validation to prove its accuracy across different demographics, languages, and regional health profiles in India. This is crucial to ensure it does not misdiagnose conditions that present differently in various populations or is not biased by being trained primarily on urban patient data.
 2. **Accountability and Oversight:** The Act would mandate that any diagnostic suggestion generated by the AI must be reviewed and confirmed by a qualified healthcare professional before any treatment is prescribed. The system must maintain a clear audit trail of the AI's suggestion and the final human decision, establishing clear accountability.
 3. **DPDP Compliance:** The collection of symptom data from patients would be governed by the DPDP Act, requiring clear notice and consent. The AI Act would ensure that the purpose for which the data is used is strictly limited to providing the requested healthcare service, preventing misuse for other purposes like marketing health products.

This dual-law system creates a comprehensive governance structure. The DPDP Act would protect the patient's rights over their data (the fuel), while the AI Act would ensure the safety, efficacy, and fairness of the algorithmic system itself (the engine). This complementary relationship is essential for building a trustworthy and effective AI-powered healthcare ecosystem in India.

Chapter 9: Conclusion and Policy Recommendations

9.1 Summary of the Argument

This paper has argued for the necessity of a forward-thinking and contextually aware legislative framework to govern the use of artificial intelligence in India's healthcare sector. The analysis began by establishing the twin imperatives of the modern digital age: the fundamental right to data protection and the transformative potential of AI-driven efficiency. It demonstrated that while these two forces are often in tension with privacy principles like data minimization conflicting with AI's need for vast datasets - a carefully crafted regulatory approach can harmonize them.

A central thesis of this paper is that the development of high-quality, ethical AI is not merely a technological challenge but is deeply intertwined with the quality of a nation's human capital. By investing in public health and education, a nation not only improves the well-being of its citizens but also cultivates the skilled innovators and generates the high-fidelity, unbiased data that are prerequisites for building trustworthy AI systems. This reframes public health expenditure as a strategic investment in technological and economic capacity.

Through a comparative analysis of the European Union's comprehensive, rights-centric model and the United States' flexible, market-driven approach, key lessons for India were identified. The EU's risk-based classification offers a structured way to tailor regulation, while the U.S. model's emphasis on adaptability highlights the need to avoid stifling innovation. The optimal path for India is not to wholly adopt either model but to forge a hybrid approach.

This paper culminates in a proposed framework for a sector-specific AI in Healthcare Act for India. This Act would not operate in a vacuum but would be built upon the foundational principles of the Digital Personal Data Protection (DPDP) Act, 2023. By creating a dual-regulatory system where the DPDP Act governs the data and the AI Act governs the algorithm, India can address the full spectrum of risks, from privacy violations to clinical safety and algorithmic bias. The proposed framework, featuring a tiered risk classification, regulatory sandboxes, and a mandate for human oversight, is designed to balance robust protection with the flexibility needed to harness AI's immense potential to solve India's unique and pressing healthcare challenges.

9.2 Actionable AI Act Framework

Based on the comprehensive analysis conducted, the following actionable policy recommendations are proposed for Indian legislators and regulatory bodies to create a responsible, equitable, and innovative AI ecosystem for healthcare:

RECITALS

Whereas, artificial intelligence (AI) offers transformative potential for healthcare by augmenting capacity, improving diagnoses, and personalizing treatment; the Government of India seeks to harness these benefits while safeguarding patient safety, equity, and public health.

Whereas, the Medical Device Rules, 2017 (as amended) explicitly treat software and algorithms used for medical purposes as medical devices [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113], and AI-driven diagnostics and decision-support tools often require similar oversight as conventional medical devices.

Whereas, the Digital Personal Data Protection Act, 2023 (DPDP Act) grants individuals fundamental rights over their digital personal data and requires explicit, informed consent for processing sensitive health data; this Act must align with the DPDP Act's principles of consent, purpose limitation, data minimization, and security.

Whereas, global best practices in AI regulation include risk-based frameworks: the United States FDA regulates AI-enabled medical software primarily through device classes (Class I–III) and voluntary pilots, whereas the European Union's AI Act employs a stratified risk scheme and bans certain categories of AI [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113]. The Government of India adopts a calibrated approach, imposing more stringent controls than the FDA but granting greater flexibility than the EU's most prohibitive rules.

Whereas, protected health information requires robust privacy and security measures; healthcare organizations must implement technical and organizational safeguards (e.g. encryption, access controls, zero-trust architectures and Software Bills of Materials) to prevent breaches and enable incident response. Software Bills of Materials (SBOMs) provide transparency of software components and are required for medical devices under U.S. FDA guidance [[HealthTech Magazine](#), Candia 2023, *5 Questions About SBOMs for Healthcare Organizations*, 22].

Whereas, transparency and explainability are essential: AI systems should be accompanied by documentation ("model cards") describing their design, training data, and performance [[IAPP Article](#), Desai 2023, *5 things to know about AI model cards*, 41], and users (clinicians and patients) must be informed when a decision or content

is AI-generated. High-risk systems must provide decision logs and user-friendly explanations of outputs, in accordance with ethical and legal requirements for accountability.

Whereas, an expert regulatory authority – the Indian AI Regulatory Authority (IARA) – is needed within the Ministry of Health and Family Welfare to oversee AI in healthcare, including rulemaking, conformity assessments, post-market surveillance, and innovation support. Regulated “sandboxes” and pilot programs can facilitate controlled testing of novel AI technologies under IARA supervision, as envisioned in recent international practices [[IAPP Article](#), Sentinella 2025, *How different jurisdictions approach AI regulatory sandboxes*, 144].

Whereas, clear liability rules are required to address harms from AI: responsibility may attach jointly to developers, deployers, and healthcare professionals, and providers of high-risk AI should carry professional liability insurance. The governance framework must make clear that AI tools augment but do not replace medical judgment.

Now, therefore, be it enacted by Parliament in the Seventy-sixth Year of the Republic of India as follows:

Article 1 – Short Title and Commencement

1. This Act may be called the Indian Artificial Intelligence (Healthcare) Act, 2025.
2. It shall come into force on such date as the Central Government may, by notification in the Official Gazette, appoint.

Article 2 – Scope

1. This Act applies to all AI systems used in healthcare contexts within India, including software, algorithms, and related systems used for diagnostics, treatment, monitoring, administration, and related health services, whether developed in India or abroad, if placed on the market or put into service in India.
2. This Act supplements and operates alongside the Digital Personal Data Protection Act, 2023, the Clinical Establishments (Registration and Regulation) Act, 2010, the Information Technology Act, 2000, the Drugs and Cosmetics Act, 1940 (including the Medical Device Rules), and other relevant laws. In case of conflict, the more stringent requirement shall prevail.

Article 3 – Definitions

For the purposes of this Act:

(a) **“AI system”** means any software or hardware system that, using techniques such as machine learning, logic- and knowledge-based approaches, or inference, is designed to operate with varying levels of autonomy and may adapt after deployment, and that for explicit or implicit objectives infers from input how to generate outputs such as predictions, content, recommendations, or decisions. This includes computer vision, natural language processing, data analytics, expert systems, and robotics used in healthcare.

(b) **“Healthcare provider”** means any person or entity (including hospitals, clinics, laboratories, telemedicine providers, and individual practitioners) that offers healthcare services, diagnoses, treatments, or medical advice to patients under applicable law.

(c) **“Patient data”** or **“personal health data”** means any information in digital form about an individual’s health or medical history, obtained from healthcare providers, devices, or patients themselves. Patient data is treated as sensitive personal data under the DPDP Act; its processing requires special safeguards.

(d) **“High-risk healthcare AI system”** means an AI system whose use in healthcare could significantly affect patient health or safety. In particular, this includes any AI system used for clinical diagnosis, treatment planning or decision support, critical monitoring, or life-critical interventions, or any AI that functions as a safety component of a regulated medical device. As a rule, any AI used in software qualifying as a Class IIa, IIb or III medical device under the Drugs and Cosmetics Act (Medical Device Rules, 2017) is high-risk [[IQVIA Article](#), King 2024, *The Future of AI in Medical Devices: FDA Guidelines and Global Perspectives*, 80]. Example: AI for interpreting radiology or pathology images to diagnose cancer; AI for real-time monitoring of ICU patients.

(e) **“Limited-risk healthcare AI system”** means an AI system with potential impact on health that is less than high-risk. These systems support healthcare processes but are not safety-critical. Example: AI-driven appointment scheduling, billing systems, non-critical triage or patient reminder chatbots, or moderate diagnostics used with human supervision [[Trail-ML Article](#), *EU AI Act: How risk is classified*, 152].

(f) **“Minimal-risk healthcare AI system”** means an AI system with negligible impact on patient health or rights. These systems provide administrative or informational functions and do not entail significant safety concerns. Example: AI used for appointment reminders, generic health information retrieval, or spam filtering of health records.

(g) **“Prohibited AI practice”** means any use of AI in healthcare that is expressly banned by this Act due to risks to fundamental rights or safety. Prohibited applications include those that intentionally manipulate patient behavior without informed consent, discriminate or profile patients on sensitive attributes (e.g. race, caste, religion) for healthcare decisions, or automate life-critical decisions without meaningful human oversight [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113]. Example: Social scoring of patients by insurance companies, or unconscious patient manipulation by AI.

(h) **“Developer”** (or **“Provider”**) means any person or organization that develops or modifies an AI system and places it on the market or puts it into service under its own name or trademark.

(i) **“Deployer”** means any person or organization that uses or integrates an AI system in healthcare under its authority, including healthcare institutions and clinicians, except when used by individuals in personal, non-professional contexts.

(j) **“Authorized representative”** means a person or entity in India who has received a written mandate from a foreign developer to perform on its behalf the obligations of this Act (including conformity assessment).

(k) **“Conformity assessment”** means the procedure to demonstrate that an AI system (especially high-risk) complies with the requirements of this Act and applicable standards.

(l) **“Post-market monitoring”** means the obligation of developers to collect and review experience gained from their AI systems after deployment, including adverse events or performance degradation, and to take corrective actions as needed.

Article 4 – Risk Classification

1. AI systems subject to this Act are divided into four risk tiers: **Prohibited, High-Risk, Limited-Risk,** and **Minimal-Risk** (as defined in Article 3). Obligations on developers and deployers vary by tier.
2. **Prohibited Tier:** AI applications described in Article 3(g) are banned. Any person placing on the market or using a prohibited AI system commits an offense.
3. **High-Risk Tier:** AI systems in healthcare that meet the definition in Article 3(d) are high-risk. High-risk systems require the most stringent requirements (see Articles 5–9). Examples include AI for primary diagnosis, critical monitoring (e.g. ICU), surgical robotics, dosage calculation, or safety components of

medical devices [[IQVIA Article](#), King 2024, *The Future of AI in Medical Devices: FDA Guidelines and Global Perspectives*, 80].

4. **Limited-Risk Tier:** AI systems with moderate impact (Article 3(e)) are limited-risk. Such systems must meet baseline transparency and accountability obligations but are exempt from full conformity assessment. Example: patient triage tools or non-critical diagnostic support. Deployers of limited risk systems must inform users when AI is used and maintain logs of usage (Articles 8–9).
5. **Minimal-Risk Tier:** AI systems with negligible impact (Article 3(f)) are minimal-risk. These systems are subject only to general principles (data protection, cybersecurity, ethics). There are no premarket authorization requirements for minimal-risk systems, but developers should follow voluntary codes of conduct (e.g. human oversight, fairness) [[Trail-ML Article](#), *EU AI Act: How risk is classified*, 152].

Article 5 – Pre-Market Obligations and Conformity Assessment

1. **Clinical Validation:** Developers of high-risk healthcare AI systems must conduct adequate premarket testing and clinical validation on data representative of the Indian population. This includes prospective or retrospective studies demonstrating the system’s safety and performance across age, gender, region, and other relevant demographics [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113]. Validation protocols should follow established medical device standards.
2. **Documentation and Model Cards:** Developers must prepare detailed technical documentation and model cards for each high- and limited-risk AI system. This documentation shall describe the system architecture, training data (including sources and demographic coverage), intended use, and performance metrics. Documentation must be kept up to date with each new version.
3. **Conformity Assessment (High-Risk):** High-risk AI systems must undergo a third-party conformity assessment by an IARA-designated body before being placed on the market or put into service. The assessment verifies compliance with this Act’s requirements (Articles 6–9) and relevant standards (see Annexes). The successful outcome must be documented in a Declaration of Conformity.
4. **AI Management System Certification:** Organizations that develop or deploy high-risk AI must establish an AI management system and certify it to ISO/IEC 42001:2023 (or an equivalent international standard) within 24 months of this Act’s commencement [[ISO/IEC 42001:2023 Standard](#), *Information technology — Artificial intelligence — Management system*, 81]. ISO/IEC 42001 provides requirements for responsible AI governance and risk management across the development lifecycle.
5. **Pre-Certification Option (“Pre-Cert”):** A developer with a proven track record of quality and compliance (as determined by IARA) may apply for a streamlined “pre-certification” pathway. Such developers can receive conditional approval for new AI products subject to ongoing quality monitoring, similar to the U.S. FDA Pre-Cert Program. IARA may require additional post-market conditions in lieu of full pre-market review for pre-certified entities.
6. **Transparency and Labeling (Limited-Risk and Above):** All limited- and high-risk AI systems must carry clear labeling or disclosures. In clinical settings, any decision or recommendation originating from an AI system must be explicitly noted in patient records and explained to treating physicians. Where an AI system directly interacts with patients (e.g. chatbots, symptom checkers), the system must inform the user that they are interacting with an AI and not a human provider.
7. **Security Requirements:** All AI systems (regardless of risk tier) must comply with state-of-the-art cybersecurity best practices. Specifically, developers must:
8. Implement **Zero Trust** principles in system design, ensuring continuous authentication and authorization for data access.
9. Maintain an up-to-date **Software Bill of Materials (SBOM)** for each AI system version, listing all software components and libraries [[HealthTech Magazine](#), Candia 2023, *5 Questions About SBOMs for Healthcare Organizations*, 22]. The SBOM shall accompany the product and be shared with IARA and purchasers to facilitate vulnerability management.
10. Apply encryption, access controls, and multi-factor authentication to protect patient data as per Section 8.

11. Immediately notify IARA and affected parties of any security breach or vulnerability that could compromise AI system safety or data integrity.

Article 6 – Data Protection and Privacy

1. **Alignment with DPDP Act:** All processing of personal data by AI systems (including training, inference, and logging) shall comply with the Digital Personal Data Protection Act, 2023. Healthcare entities and AI developers are Data Fiduciaries under that Act. This includes honoring rights of access, correction, and erasure for data principals (patients).
2. **Consent:** Explicit, informed consent must be obtained from each patient before using their personal health data in an AI system, except where anonymized data is used purely for statistical public health purposes. Consent forms must explain the purpose of the AI, potential risks, and any automated decision-making involved. Consent for high-risk AI uses shall be separate and more detailed.
3. **Anonymization/Pseudonymization:** Patient data used for training or improving AI must be deidentified in accordance with DPDP guidelines. When datasets are not fully anonymized, robust pseudonymization techniques and strict access controls must be applied. Use of personal data must be minimized to the extent necessary for the AI's intended purpose.
4. **Data Minimization and Purpose Limitation:** AI systems shall be designed to collect and retain only the minimum patient data needed for their specified healthcare function. Secondary use of health data (e.g. for unrelated research or marketing) is prohibited without fresh patient consent. Data retention periods shall be limited to what is strictly necessary, with secure disposal protocols thereafter.
5. **Audit Trail and Accountability:** AI systems must maintain detailed audit logs of data processing activities. The logs shall record what patient data was input, when and by whom, and what outputs or decisions were generated. These logs must be tamper-evident and retained for a prescribed period to enable audits by IARA or patient requests. The duty to maintain an audit trail extends to both developers and deployers when operating high-risk systems.

Article 7 – Transparency and Explainability

1. **Model Cards and Documentation:** For each high and limited-risk AI system, providers must publish a model card summarizing its intended use, performance metrics on different demographic groups, known limitations, and training data characteristics [[IAPP Article](#), Desai 2023, *5 things to know about AI model cards*, 41]. This is to promote transparency and allow clinicians/regulators to assess suitability. Model cards shall be in a standardized format approved by IARA.
2. **Decision Logs:** High-risk AI systems must maintain “decision logs” recording each instance of inference or recommendation, including input parameters, model version, confidence scores, and rationale. These logs shall be made available in human-readable form to clinicians auditing treatment decisions or in the event of adverse outcomes. Maintaining decision logs enhances accountability and safety.
3. **Explainable AI:** High-risk AI systems must be designed with explainability in mind. Providers shall implement state-of-the-art explanation techniques so that outputs can be interpreted by qualified professionals. When deployed, the AI's predictions or suggestions must be accompanied by information on which factors influenced the result. This is necessary to ensure clinicians and patients understand AI-driven decisions [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113].
4. **User Notification:** Any use of AI in patient-facing interactions must be clearly labeled. Patient portals, mobile apps, or medical devices employing AI shall display a statement such as “Powered by AI” when the patient is receiving AI-generated advice or analysis. Generated medical content (e.g. health summaries, images) must be marked as AI-generated to avoid confusion.

5. **Deepfake and Synthetic Media:** Any synthetic media or “deepfake” outputs used for medical education or public health messaging must be explicitly labeled as such. Providers of AI imaging or audio synthesis technologies must implement safeguards to prevent misuse or misleading outputs.

Article 8 – Post-Market Surveillance and Reporting

1. **Vigilance and Incident Reporting:** Developers of high-risk AI systems must establish a post-market surveillance system. They must monitor real-world performance, collect reports of adverse events or malfunctions, and file periodic vigilance reports to IARA. Any serious incident (death, irreversible harm, or system malfunction causing significant treatment delay) must be reported to IARA within 7 days of awareness.
2. **Cooperation with Regulators:** Deployers (healthcare providers) must report to developers and IARA any adverse events or near-misses involving AI. They must preserve relevant logs and data for investigations. Both developers and deployers shall cooperate with IARA audits and provide access to data or software as needed.
3. **Performance Reassessment:** If significant changes occur (e.g. model retraining, environment shift), developers must reassess the AI’s performance. Substantial modifications aftermarket (not foreseen in initial approval) require fresh conformity assessment and possible re-certification.
4. **Continuous Improvement:** Developers are encouraged to update AI systems to improve accuracy and safety. However, updates shall follow the same validation and documentation procedures as new products.

Article 9 – Cybersecurity and Integrity

1. **Zero-Trust Architecture:** All healthcare AI systems must employ a “zero-trust” security model, meaning that no user or component is automatically trusted. Access to patient data and system functions must require strict identity verification, least-privilege permissions, and continuous monitoring of suspicious activity. This applies to AI systems hosted on-cloud or on-premises.
2. **Supply Chain Security (SBOM):** As required by U.S. guidance, every AI system must include a comprehensive SBOM [[HealthTech Magazine](#), Candia 2023, *5 Questions About SBOMs for Healthcare Organizations*, 22]. The SBOM shall identify all third-party and open-source software components and their versions. Providers must update the SBOM with each software release and share it with healthcare institutions to facilitate vulnerability patching.
3. **Encryption and Data Security:** Patient data in transit and at rest must be encrypted with strong encryption standards. AI models and outputs involving patient data should be stored on secure infrastructure that complies with the IT Act’s reasonable security practices.
4. **Security Certifications:** High-risk AI systems should, where applicable, comply with international cybersecurity standards (e.g. ISO/IEC 27001 for information security management) and with Indian CERT-In guidelines. IARA may require specific security audits or penetration tests for particularly sensitive AI deployments.

Article 10 – Indian AI Regulatory Authority (IARA)

1. **Establishment:** The Central Government shall, within 180 days of this Act’s commencement, establish the Indian AI Regulatory Authority (IARA) as a specialized body under the Ministry of Health and Family Welfare. The IARA shall have technical, legal, and medical experts to oversee AI in healthcare.
2. **Functions and Powers:** IARA shall have the following powers, subject to the Ministry’s general direction:
 - a) **Rulemaking:** Make rules and guidelines for implementing this Act (e.g. standards for model cards, audit protocols, certification procedures).

- b) **Registration:** Maintain a registry of all high-risk AI systems placed in the Indian market, including vendor and performance details. Access to the registry may be limited to IARA and other authorized agencies.
- c) **Conformity Oversight:** Authorize and monitor conformity assessment bodies for AI (including medical device notified bodies). Approve testing laboratories for clinical validation.
- d) **Inspections and Audits:** Conduct inspections of developers and deployers to verify compliance (data handling, documentation, security). Investigators may require access to source code, training data logs, or decision logs as needed.
- e) **Enforcement:** Impose administrative sanctions (warnings, fines) for non-compliance (see Article 12). In serious cases, recommend suspension of an AI system's use. Coordinate with law enforcement for criminal violations.
- f) **Grievance Redressal:** Establish a mechanism for patients and clinicians to report harms or file complaints about AI systems. Ensure timely investigation and response to public concerns.
- g) **Standards and Codes:** Promote development of Indian and international standards for AI in healthcare. Recognize and harmonize standards (including ISO/IEC 42001, medical device standards, SBOM formats).
- h) **Innovation Sandboxes:** Operate or supervise regulatory "sandboxes" where developers can test new AI technologies under temporary relaxed requirements. IARA may grant time-limited exemptions for pilot projects under strict conditions (data protection, oversight, reporting).
- i) **International Cooperation:** Engage with foreign regulators (e.g. EU AI Office, FDA) for cross border oversight of AI tools. Consider mutual recognition of conformity assessments when appropriate.

Article 11 – Liability and Professional Accountability

1. **Joint Liability:** Liability for harm caused by a healthcare AI system shall attach to all responsible parties. Providers (developers) are liable for defects in design or data; deployers (healthcare institutions or professionals) are liable for improper use or failure to supervise. Courts may apportion liability based on fault and role. Liability insurance requirements for providers and deployers are as specified below.
2. **Medical Malpractice:** Clinical personnel using AI must exercise professional judgment. No AI output creates a defense from established medical malpractice duty. IARA clarifies that AI is a tool that augments medical care [[Indian Journal of Medical Ethics](#), Naik and Reddy 2025, *A framework for the governance of AI in healthcare for India: An exploratory model*, 113]. Physicians remain responsible for informed consent and standard of care.
3. **Mandatory Insurance:** Deployers of high-risk AI systems (including hospitals, clinics, and telemedicine platforms) must maintain malpractice insurance that covers AI-related liability. Minimum coverage limits shall be prescribed by regulation (not less than ₹10 crore per incident). Developers of high-risk AI should similarly carry product liability insurance or indemnity.
4. **Harm Compensation:** In case of injury or death attributable to an AI system, affected patients or families may seek compensation through civil remedies. The Act does not limit statutory consumer or medical tort liabilities. Alternative dispute resolution mechanisms (mediation, arbitration) for AI cases may be encouraged by IARA.

Article 12 – Enforcement and Penalties

1. **Administrative Fines:** Contravention of duties under this Act by any person (developer, deployer, medical professional, or other operator) is punishable by administrative fines, in accordance with the principle of being effective, proportionate and dissuasive. For the gravest breaches (see section 2), the fine shall be up to 4% of the total worldwide annual turnover of the offender (if a company) or ₹100 crore, whichever is higher [[GDPR Art 83\(5\)](#), 59]. Lesser violations may incur smaller fines (e.g. up to ₹10 crore or 1% turnover). (The exact schedule of penalties shall be notified.)
2. **Willful Violations:** Knowing or willful violations (such as intentionally falsifying data, intentionally breaching patient confidentiality, or developing prohibited AI) are criminal offenses. Such offenses may

be punished with imprisonment of up to five years and/or fines up to ₹50 crore, in addition to administrative penalties. (For example, willful non-consensual use of patient data or intentional obfuscation of AI decision logs is culpable misconduct.)

3. **Licensing Actions:** Persistent or severe non-compliance may result in revocation of the AI system's registration/certification, suspension of its market authorization, or banning future use. IARA may order a recall or disabling of non-compliant AI systems.
4. **Enforcement Priority:** In determining penalties, IARA shall consider factors such as harm caused, negligence versus intent, extent of violation (number of affected patients), and efforts to remediate. A percentage reduction (e.g. up to 50%) may be applied for entities that proactively disclose violations and cooperate fully (self-reporting "leniency").

Article 13 – Innovation Sandboxes and Pilot Programs

1. **Regulatory Sandboxes:** To foster innovation, IARA shall create a framework for AI Healthcare Sandboxes. Eligible entities (especially startups and research institutions) may apply to test novel AI solutions under regulatory supervision. Within a sandbox, certain requirements (such as full conformity or data localization) may be temporarily relaxed, provided risks are contained (e.g. synthetic or pseudonymized data, informed participant consent). Sandboxes will have time and scale limits; they aim to gather real-world evidence for future rule-making [[EU AI Act Article 57](#), 51].
2. **Pilot Exemptions:** IARA may grant limited exemptions or "no-action letters" for pilot deployments of promising AI tools in predefined settings (for example, a mobile health app in a rural clinic). These exemptions are conditional on strict oversight, reporting obligations, and participant safeguards. If a pilot demonstrates safety and benefit, it may inform fast-track approval of the full system.
3. **Support for Startups:** Special guidance and streamlined processes shall be available for small and medium enterprises. Sandboxes and pre-cert programs shall prioritize innovative solutions that address public health needs. IARA may fund or collaborate in demonstration projects for AI-based diagnostics and telemedicine in underserved areas.

Article 14 – Relationship with Other Laws

1. **Clinical Establishments:** AI-equipped clinical entities must also comply with the Clinical Establishments Act (2010) and its rules (e.g. registration of hospitals, minimum facility standards). AI systems that automate clinical functions do not exempt an establishment from these obligations.
2. **Drugs and Cosmetics Act:** Any AI software deemed a "medical device" under the Drugs and Cosmetics Act shall meet the requirements of that Act and the Medical Device Rules, in addition to this Act. Where Medical Device rules require clinical investigation or performance evaluation, AI devices must fulfill those requirements.
3. **Information Technology Act:** Unauthorized access or data breaches involving AI systems may attract penalties under Section 43A (compensation for breach of security practices) and Section 72A (punishment for disclosure of personal information) of the Information Technology Act, 2000. Use of AI for fraudulent or unlawful purposes may also be penalized under the IT Act's cybercrime provisions.
4. **Data Protection Act:** All provisions of the DPDP Act, 2023 apply to health data and AI processing thereof. This Act does not override patient rights to access, correction, or erasure under the DPDP Act. Any conflict shall be resolved in favor of the DPDP Act (e.g. priority of individual consent rights).
5. **Allied Legislation:** This Act shall be read in harmony with the National Health Policy, National Digital Health Mission policies, and other relevant statutes. The Central Government may issue clarifications to resolve overlaps.

Article 15 – General Provisions

1. **Entry into Force:** This Act shall become applicable nationwide on the date of its notification. The Government may phase in certain requirements (e.g. full compliance deadlines up to 2 years for some high-risk AI obligations) to allow industry adjustment.
2. **Review and Adaptation:** IARA shall review the implementation of this Act every 3 years, and the Central Government shall report to Parliament on progress. Amendments may be proposed to address new AI technologies or policy needs.
3. **Technical Annexes:** The Central Government may issue regulations or attach annexes specifying technical standards, e.g. minimal dataset representativeness criteria, performance thresholds (sensitivity, specificity) for AI tools, and formats for audit reports and model cards. Annexes attached to this Act (or future rules) are integral to its provisions.
4. **Savings:** Any AI system lawfully placed on the market or put into service before the commencement of this Act may continue to be used, subject to any transitional rules prescribed by IARA. However, if such a system undergoes substantial modification or upgrade, it must be re-classified and reassessed under this Act.

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